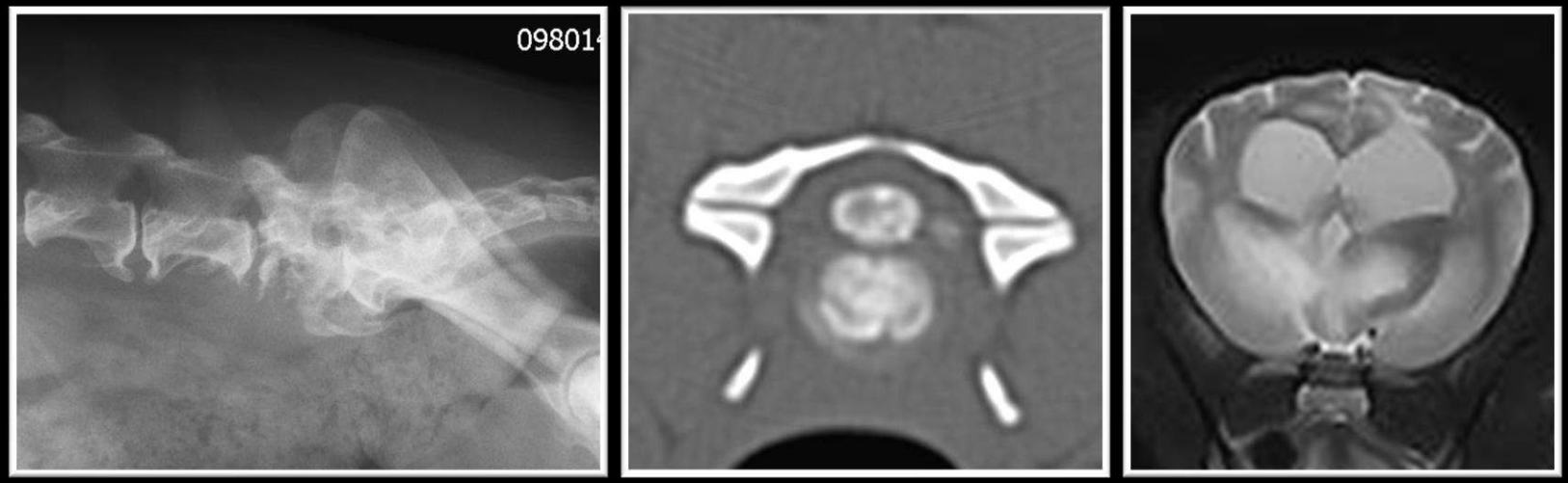
Diagnostic Imaging in Small Animal Neurology (spinal cord diseases)



國立台灣大學獸醫專業學院

張雅珮　BVM, MVM, PhD, DipECVN, DAiCVIM (Neurology)

Which one to choose?

- Radiography
- Myelography
- CT
- MRI

CNS

- Brain
- **Spinal cord**

Surroundings

- Soft tissues
- Bones

Which one to choose?

- Radiography
- Myelography
- CT
- MRI

- Which structure?
- Imaging technique
  - Advantages
  - Disadvantages
  - Limitations
  - Availability

How to achieve the accurate diagnosis?

Neuro exam ➜ Correct lesion localization

Differential diagnosis

Appropriate investigation plan

- Imaging is not everything !!!
- **Findings: significant vs. insignificant !!!**

Accurate diagnosis

Radiography

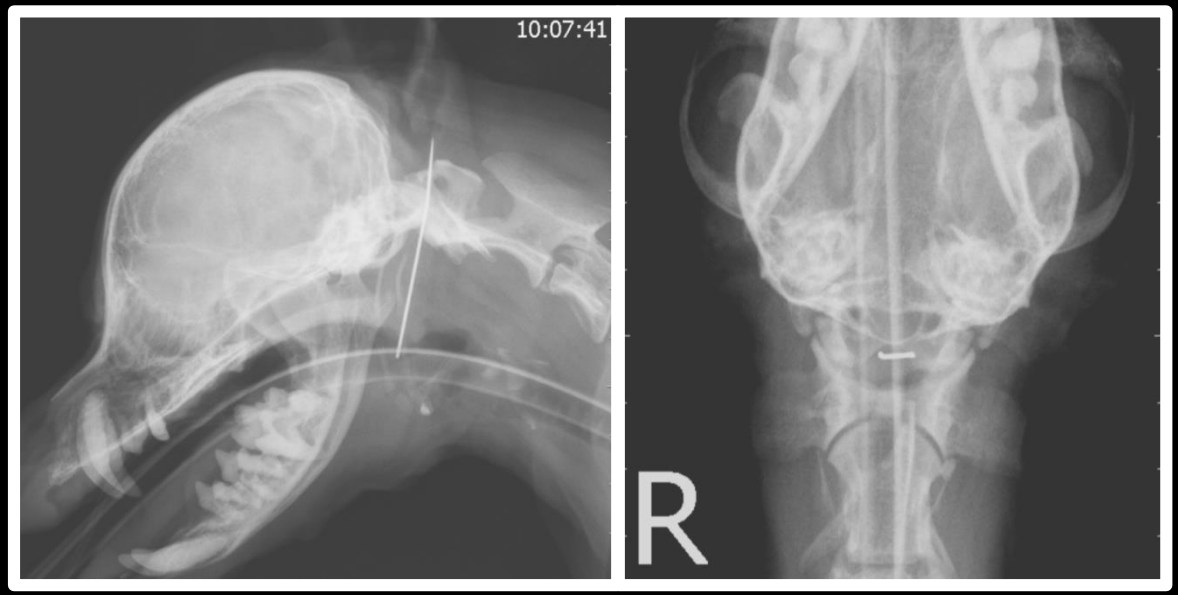


Radiography

- Head
- Spine







Vertebral anatomy

- An atlas of interpretative radiographic anatomy of the dog and cat

  Arlene Coulson and Noreen Lewis
- Textbook of veterinary diagnostic radiology

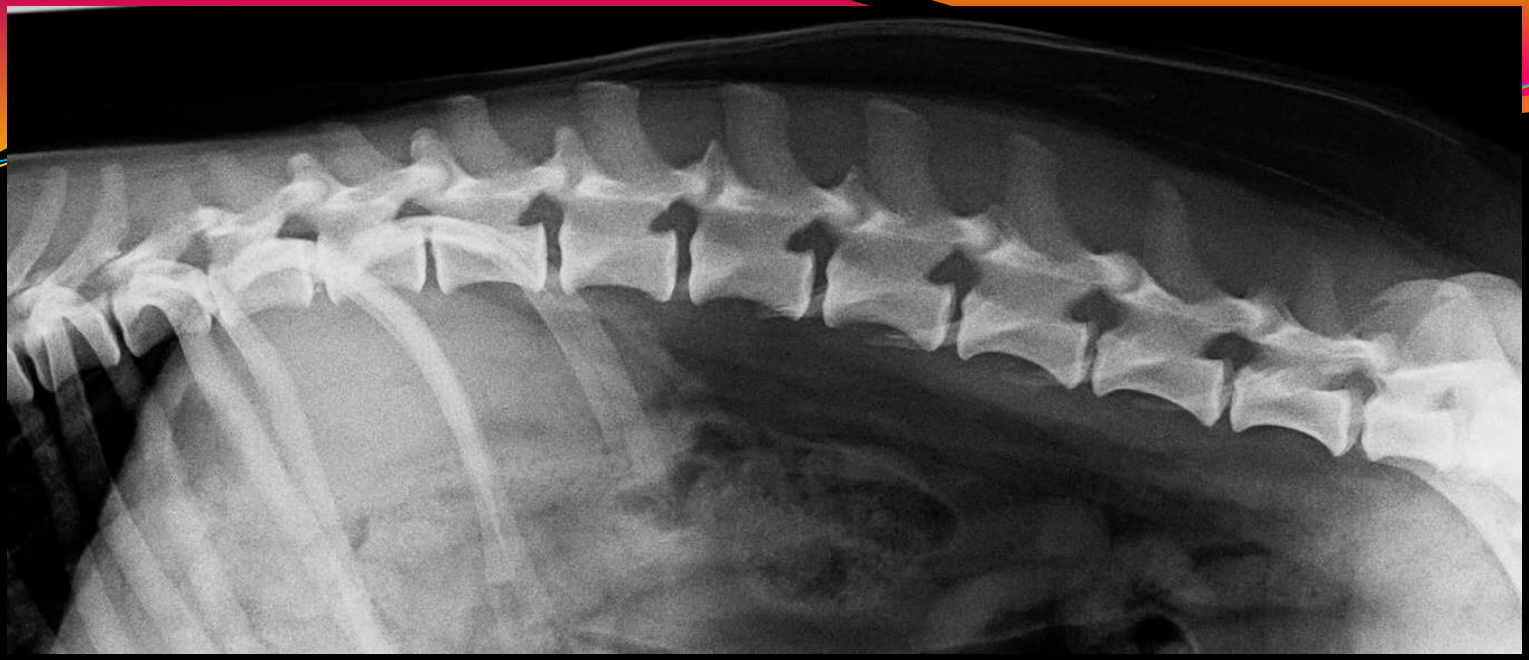
  Donald E. Thrall



















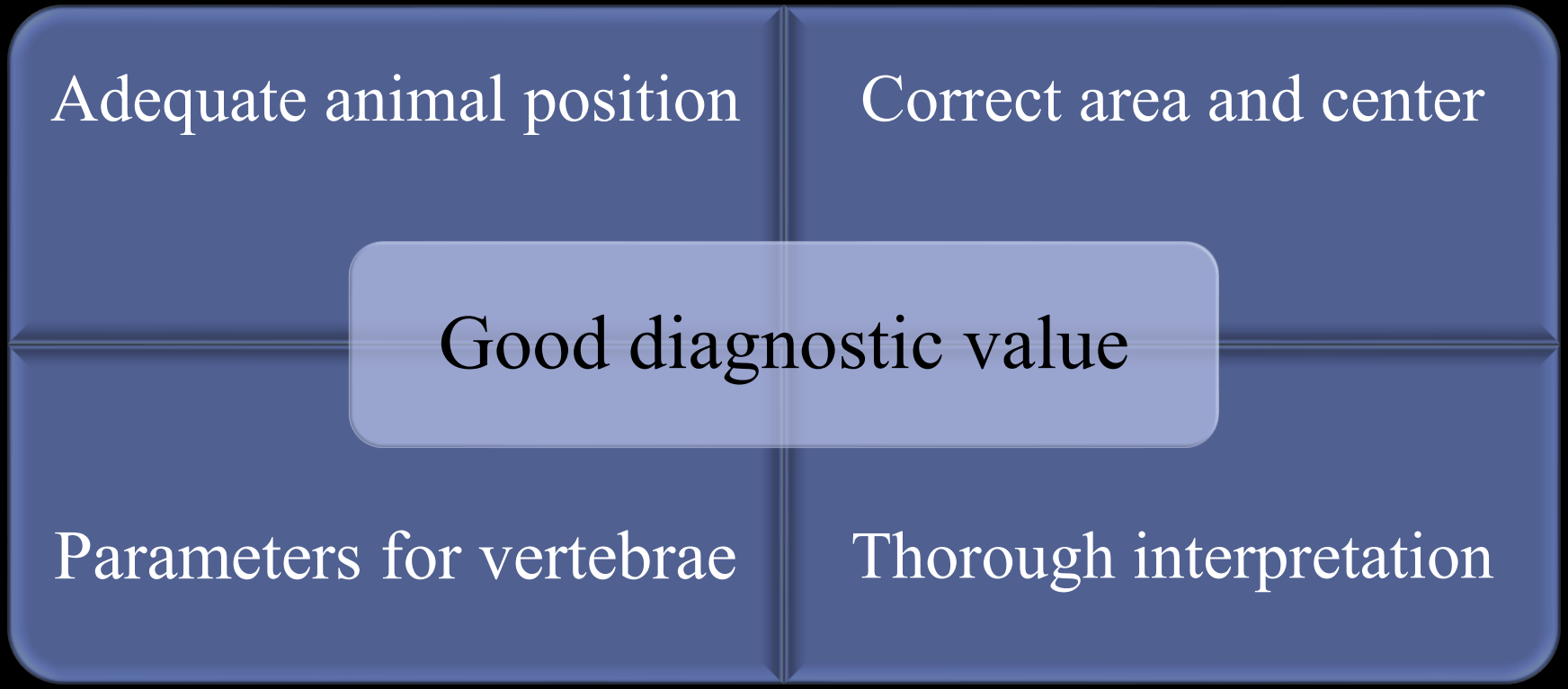






Radiography – Spine

How to make the most out of it?



Adequate animal position

- Physical restraint
- Mechanical restraint
  - Sedation or anesthesia?
- Positioning devices





Radiography - Spine





Radiography

- Lateral projection
- Ventrodorsal projection

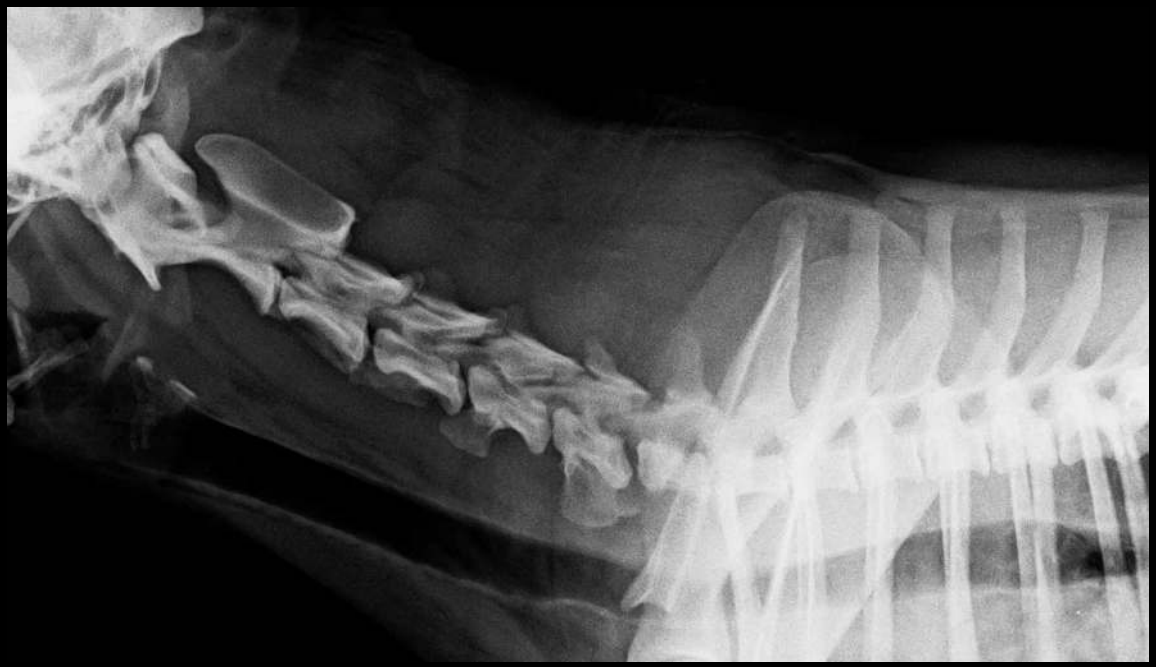
- Oblique projection
- Flexion and extension lateral projections
- Traction

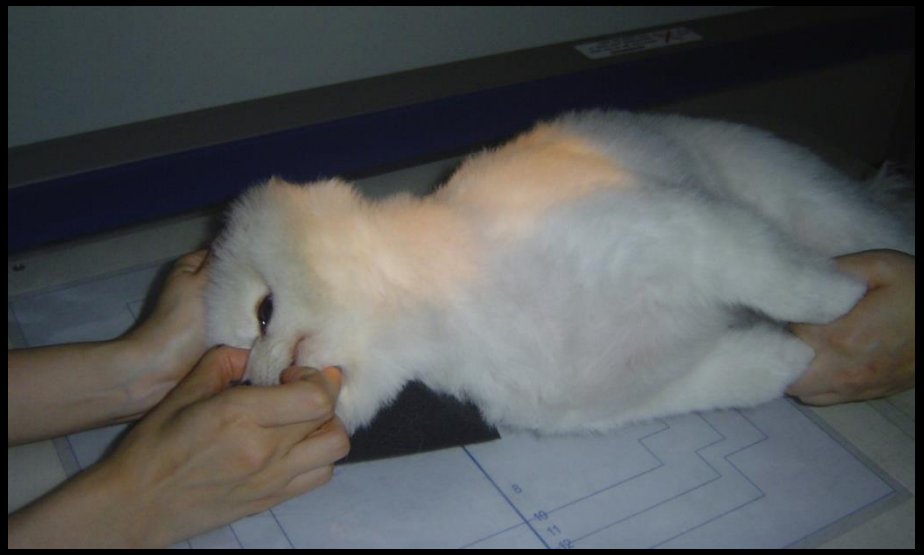
Spine – lateral projection

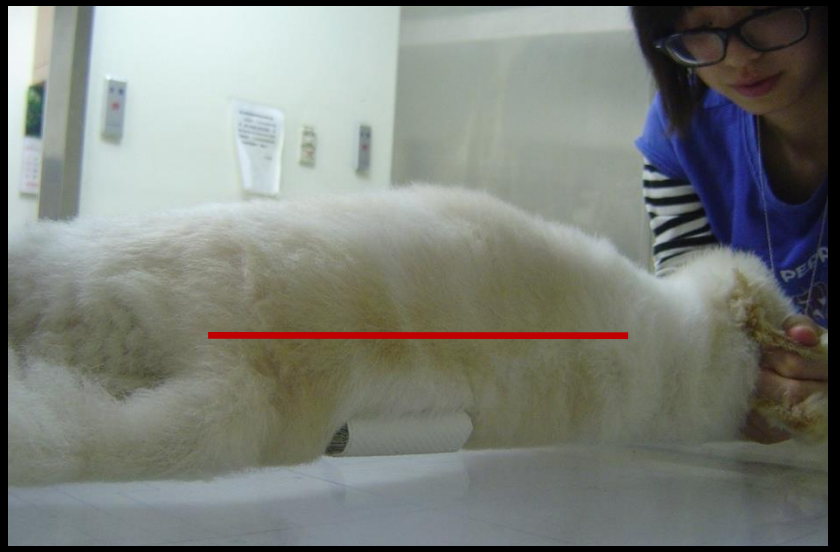
- Reducing superimposed structures
  - Cervical spine: thoracic limbs pulled caudally
  - T-L spine: thoracic limbs pulled cranially
    pelvic limbs pulled caudally
- Landmarks
  - C1, C6
  - Dorsal spinous process of thoracic vertebrae vs. sternum
  - Scapula
  - Iliac crest





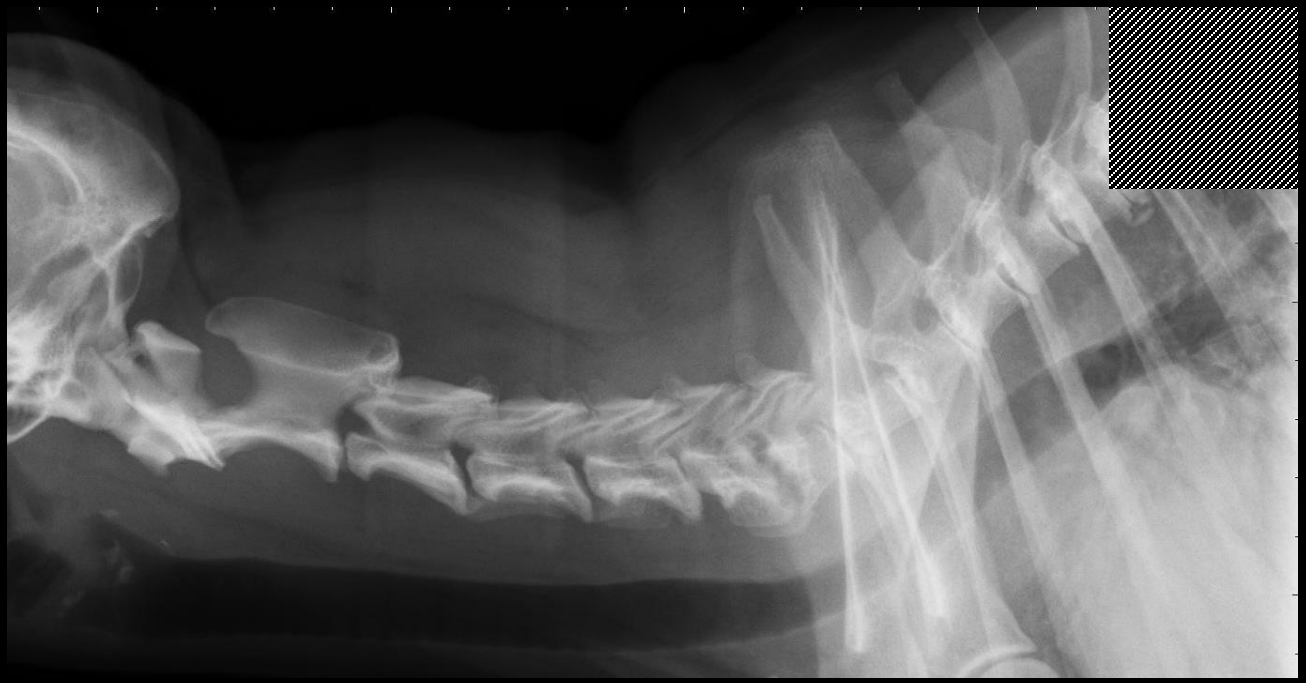



















R

Spine – VD projection

- Reducing the superimposed structures or materials
  - Endotracheal tube
  - Skull vs. C1
- Landmarks
  - C1, C6
  - Sternum
  - Shoulders, elbows, hip and stifles







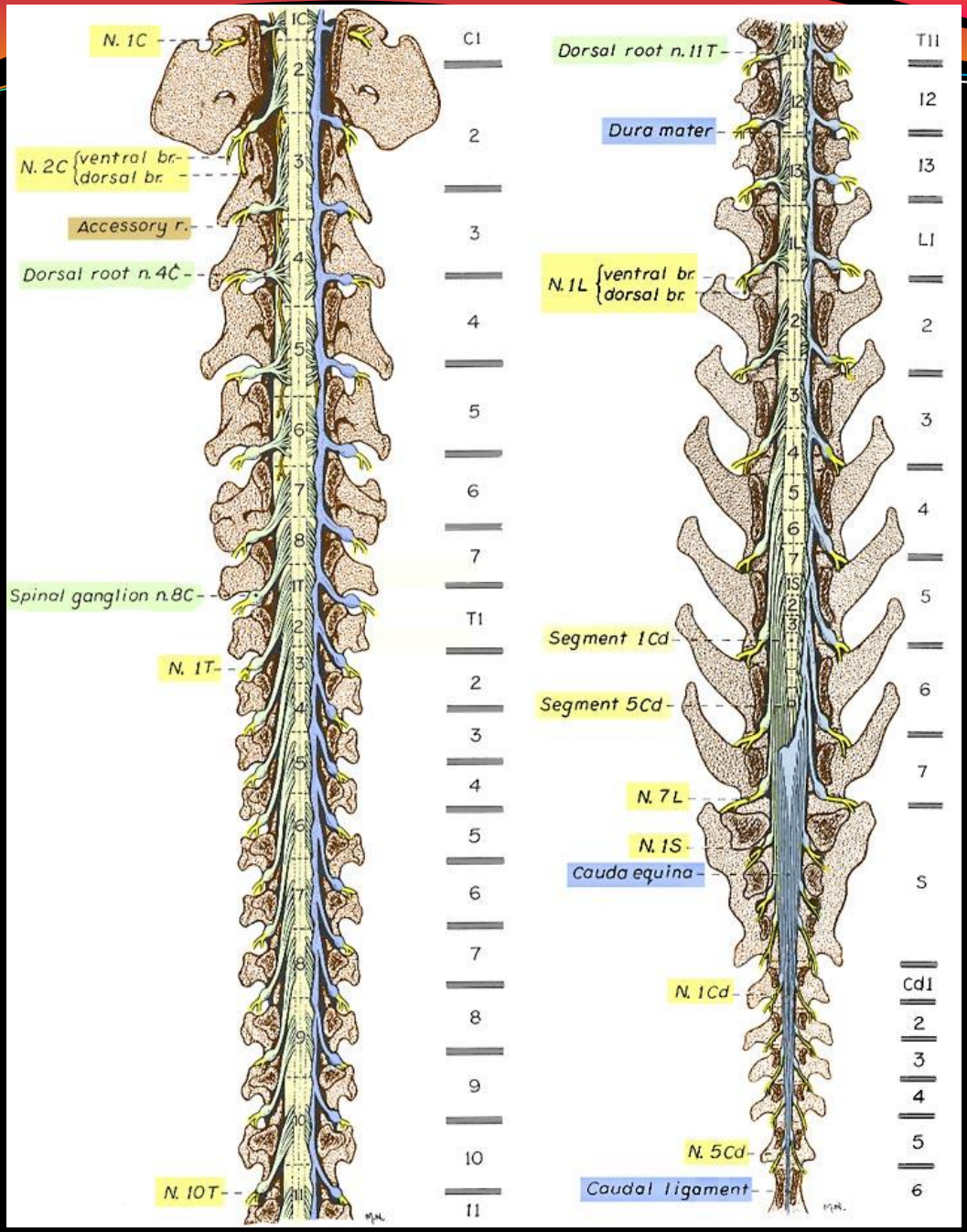




Area ?

Neurological exam → Lesion localization

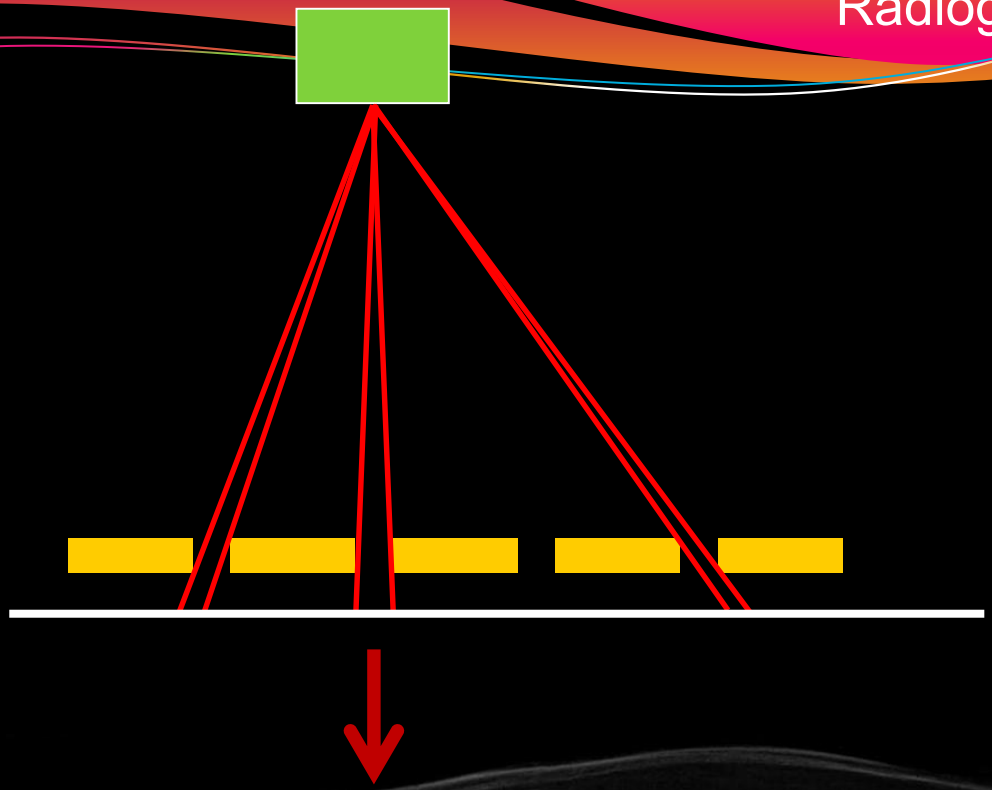
- Spinal cord segments vs. vertebral number
- Covering all areas that may cause the clinical signs

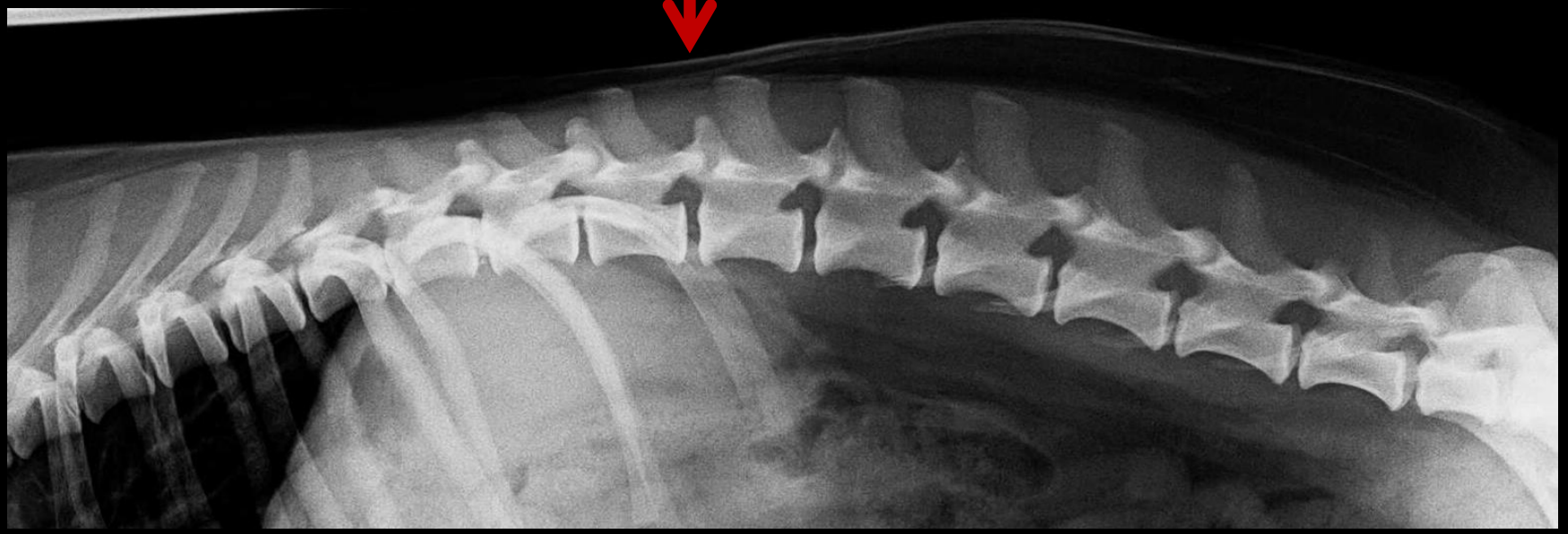


http://vanat.cvm.umn.edu/neurLab2/images/SpCord-Vert_FLAT730_colOptr.jpg

Center ?

- Disc space





Center ?

Region	Center
Cervical	C3-C4
Cervico-thoracic	C7-T1
Thoracic	T6-T7 (caudal border of scapula)
Thoraco-lumbar	T13-L1
Lumbar	L3-L4 (between the last rib and hip joint)
Lumbo-sacral	L7-S1

Thorough Interpretation

Examine the entire radiograph systemically

- Alignment of the vertebral column
- Shape of the vertebrae
- Opacity of the vertebrae
- Intervertebral disc space and foramen: size, opacity

Describe the abnormalities

Location	Margins	Number
Opacity	Size	Shape

Differential diagnosis

- VITAMIN D (DAMNIT V)

Examining the radiographs

- Patient ID
- Determine the position (projection, right or left…)
- Check…
  - Good diagnostic quality?
  - Adequate detail and contrast?
  - Proper centering?
  - Proper collimation?
  - Proper processing?

Spine – radiographic interpretation

- A SPINE
  - **A**lignment
  - **S**oft tissues
  - **P**rocesses
  - **I**nterior: vertebral canal
  - **N**erves / spinal cord: intervertebral foramen
  - **E**xterior: vertebral body, intervertebral disc

Spine – radiographic interpretation

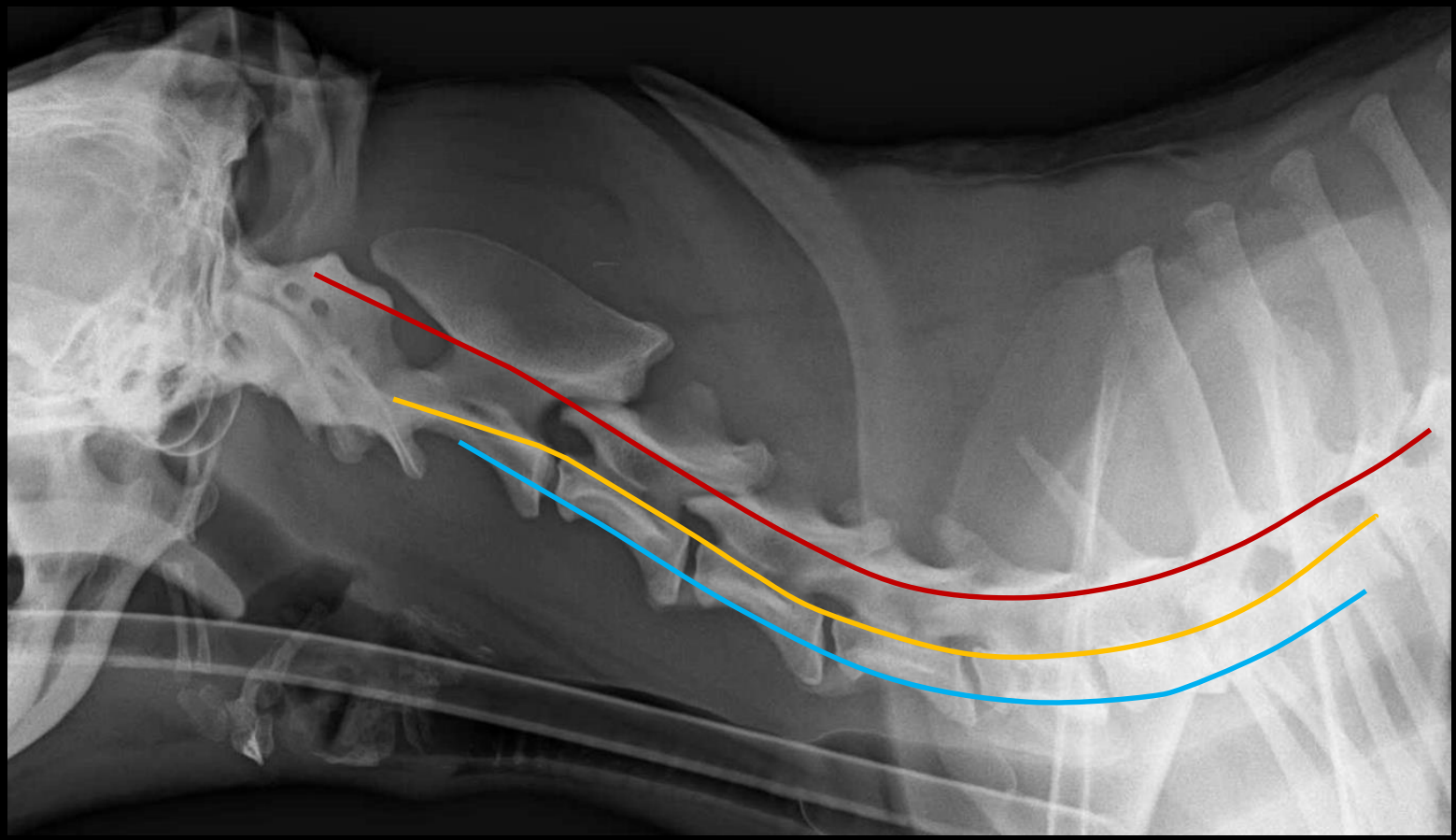
- Alignment of the vertebral column
- Shape of the vertebrae
- Opacity of the vertebrae
- Intervertebral disc space and intervertebral foramen
  - Width /size
  - Opacity

Differential diagnosis

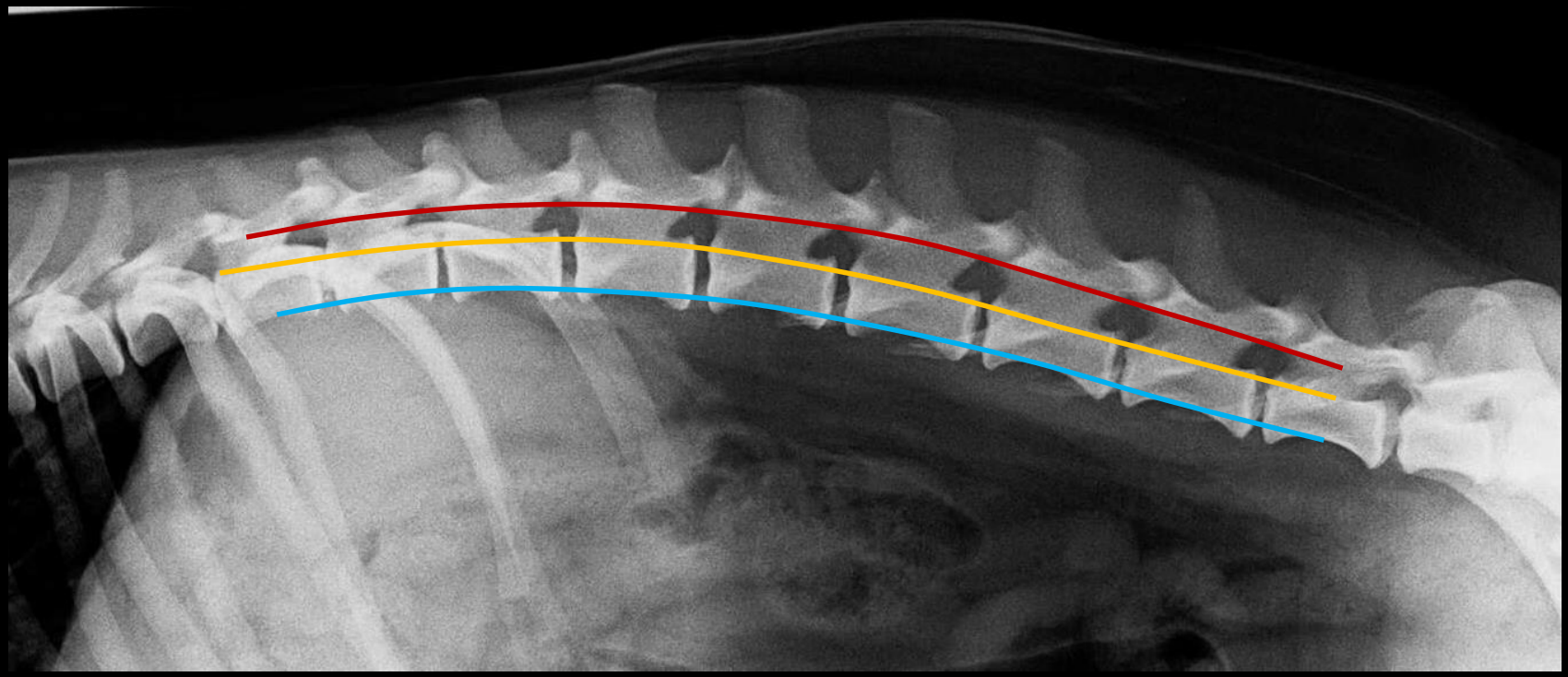
Alignment of the vertebral column

- Spinal curvature
- Traumatic fracture or luxation
- Atlanto-axial instability
- Lumbosacral instability

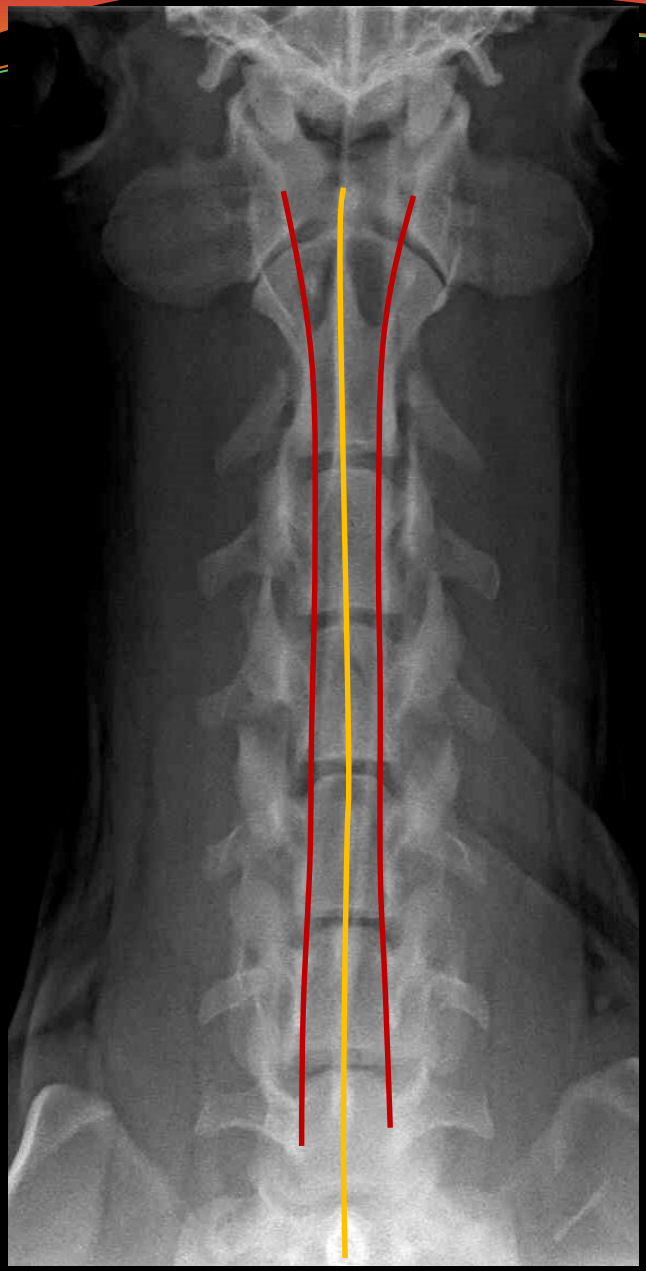
Alignment

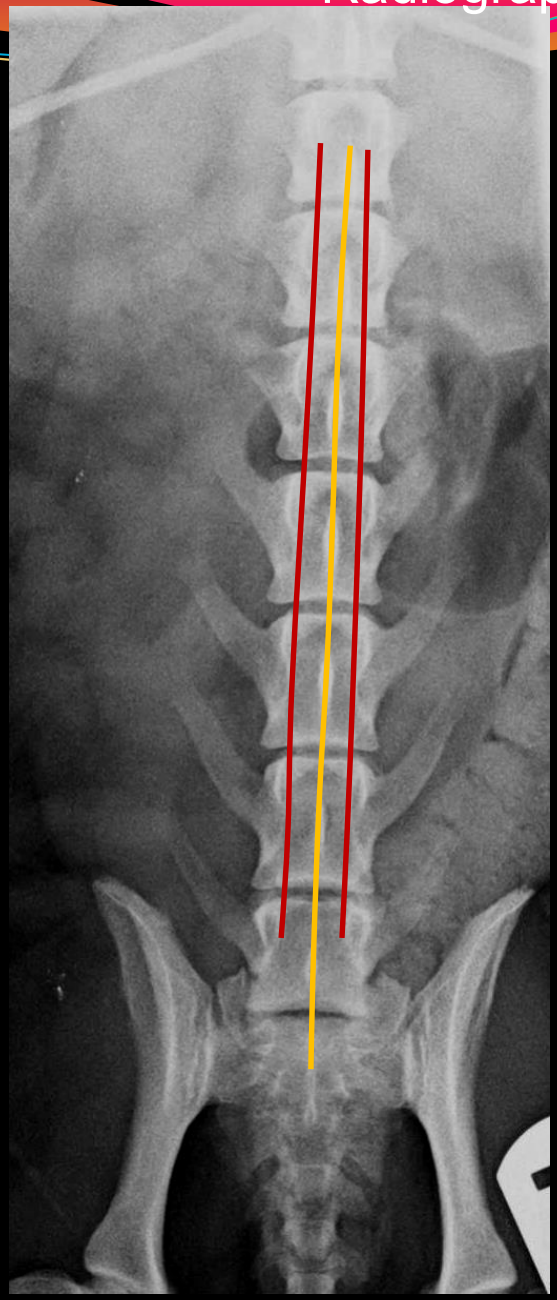


Radiography - Spine



Radiography - Spine

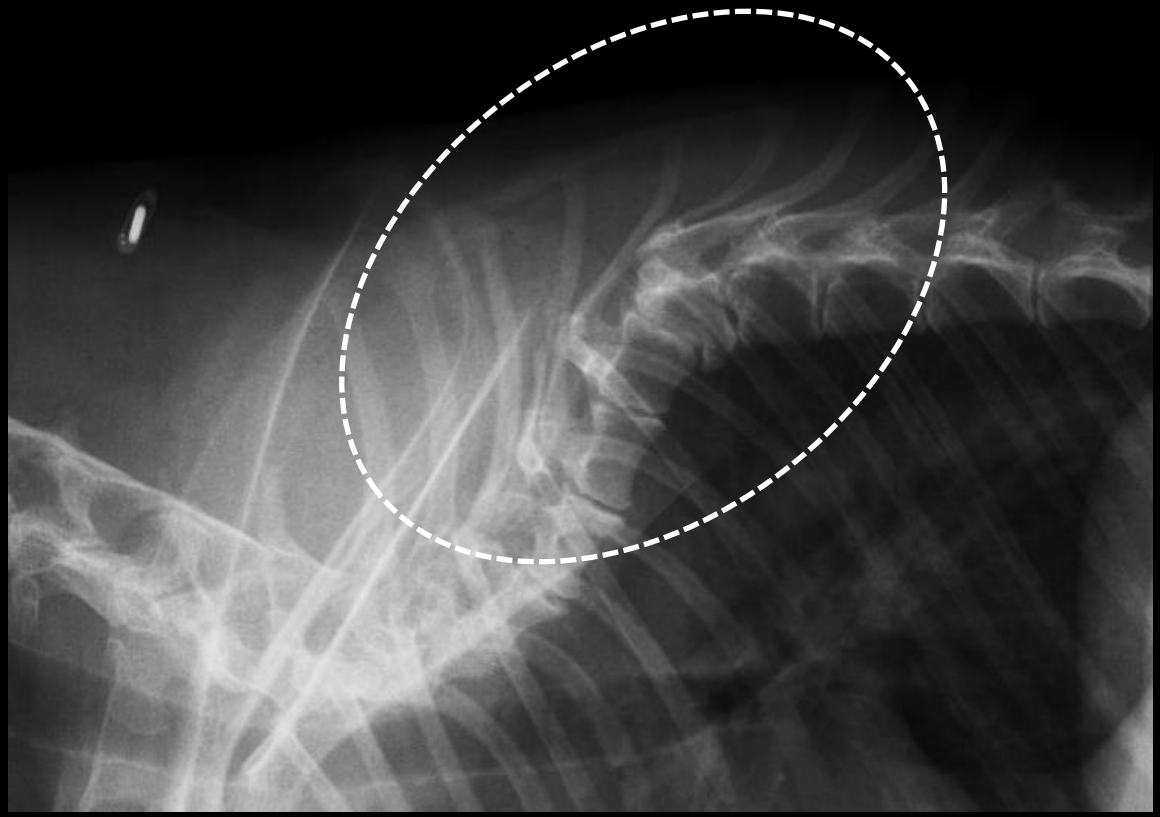




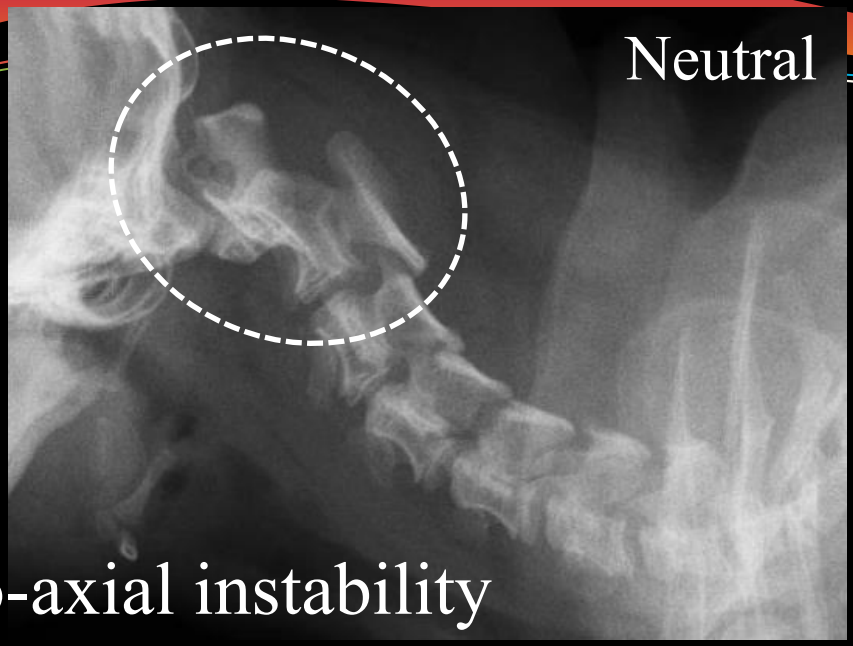
T12-T13 sub-luxation



Hemivertebrae







Atlanto-axial instability



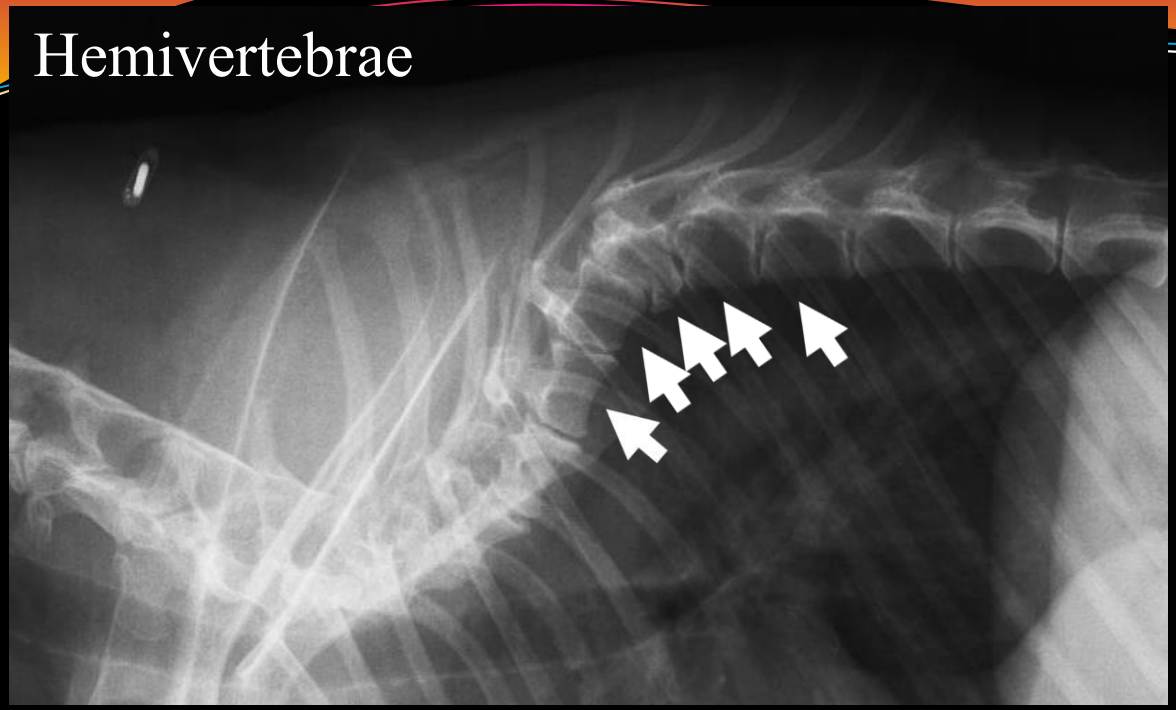


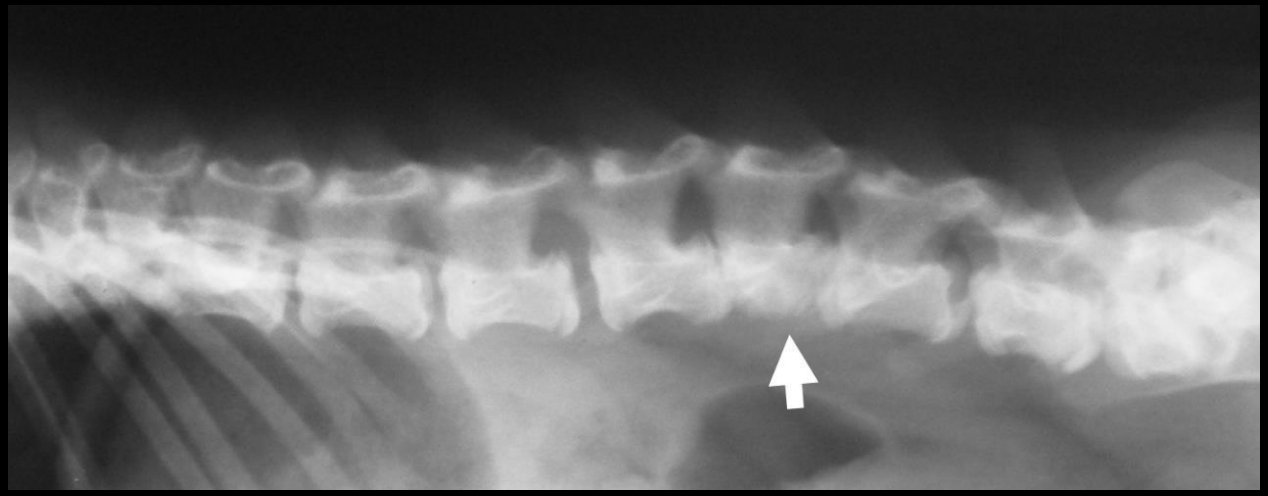
Differential diagnosis

Shape of the vertebrae

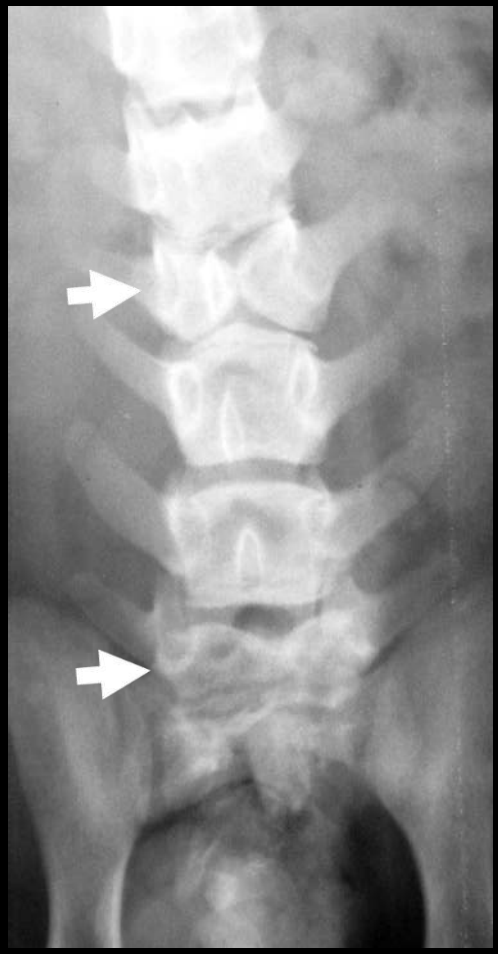
- Congenital anomaly
- Fracture
- Infection
- Neoplasia
- Spondylosis deformans
- Mucopolysaccharidosis

Hemivertebrae

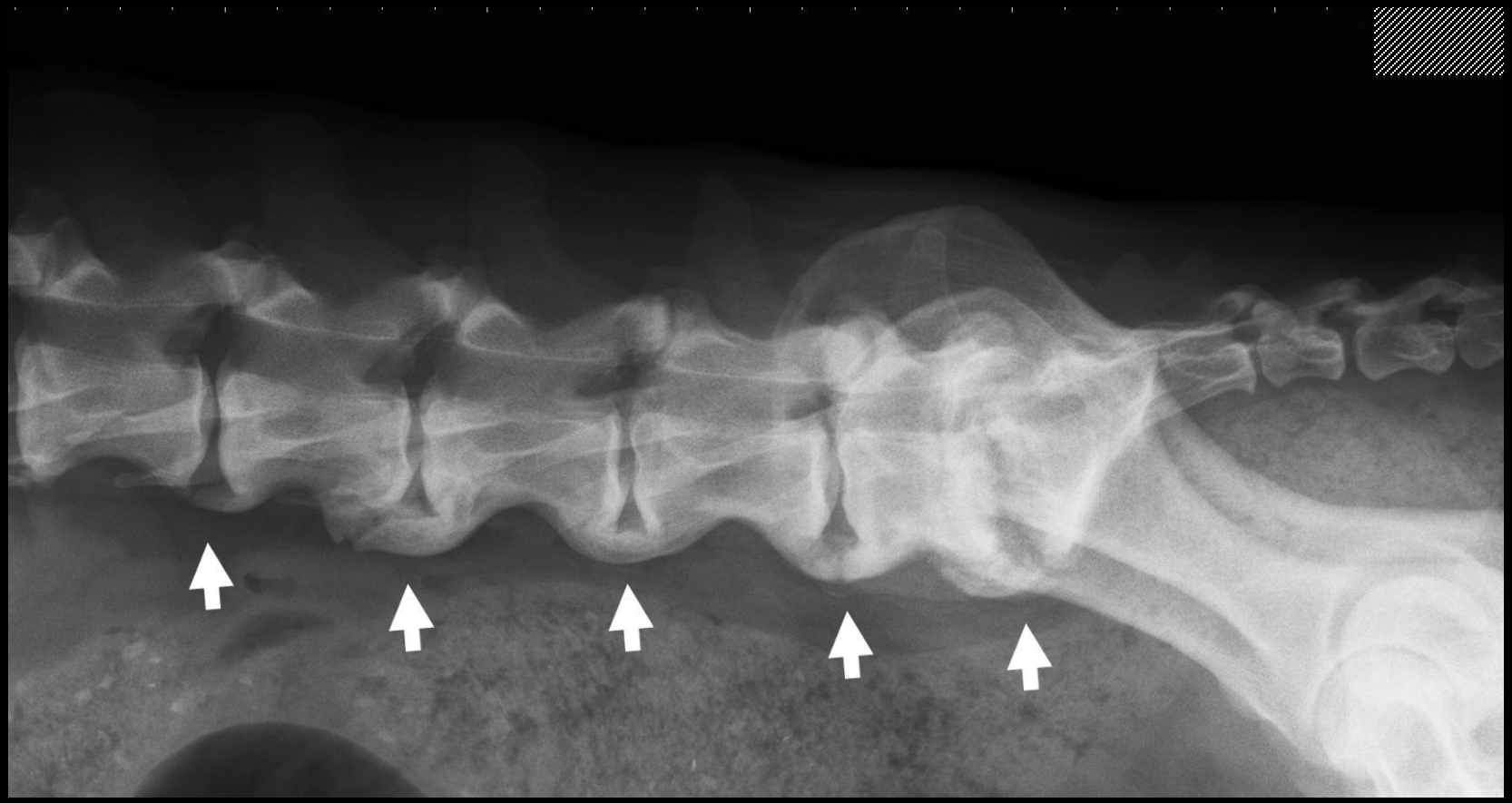


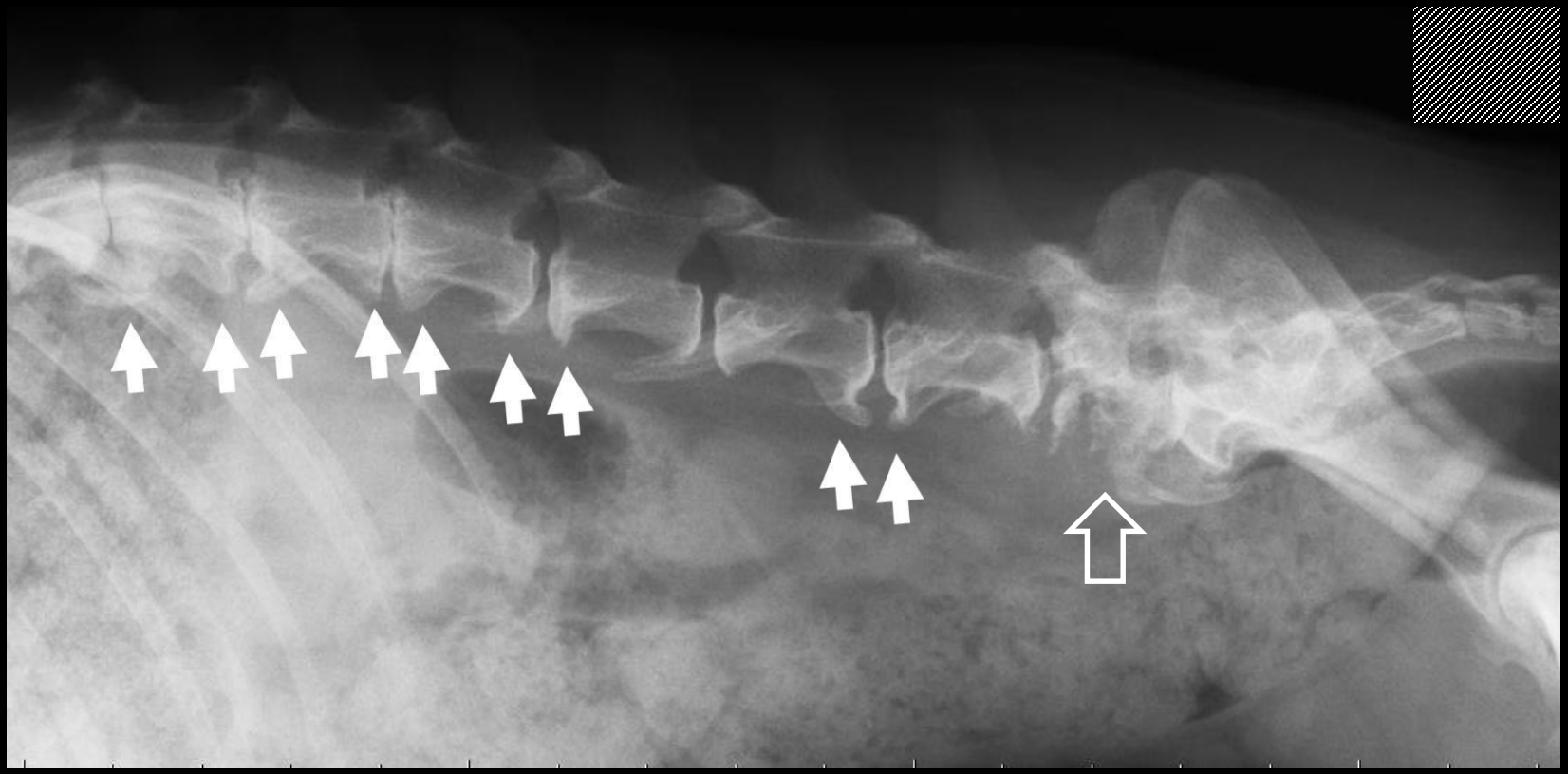


Butterfly vertebrae



Spondylosis deformans





Spondylosis deformans

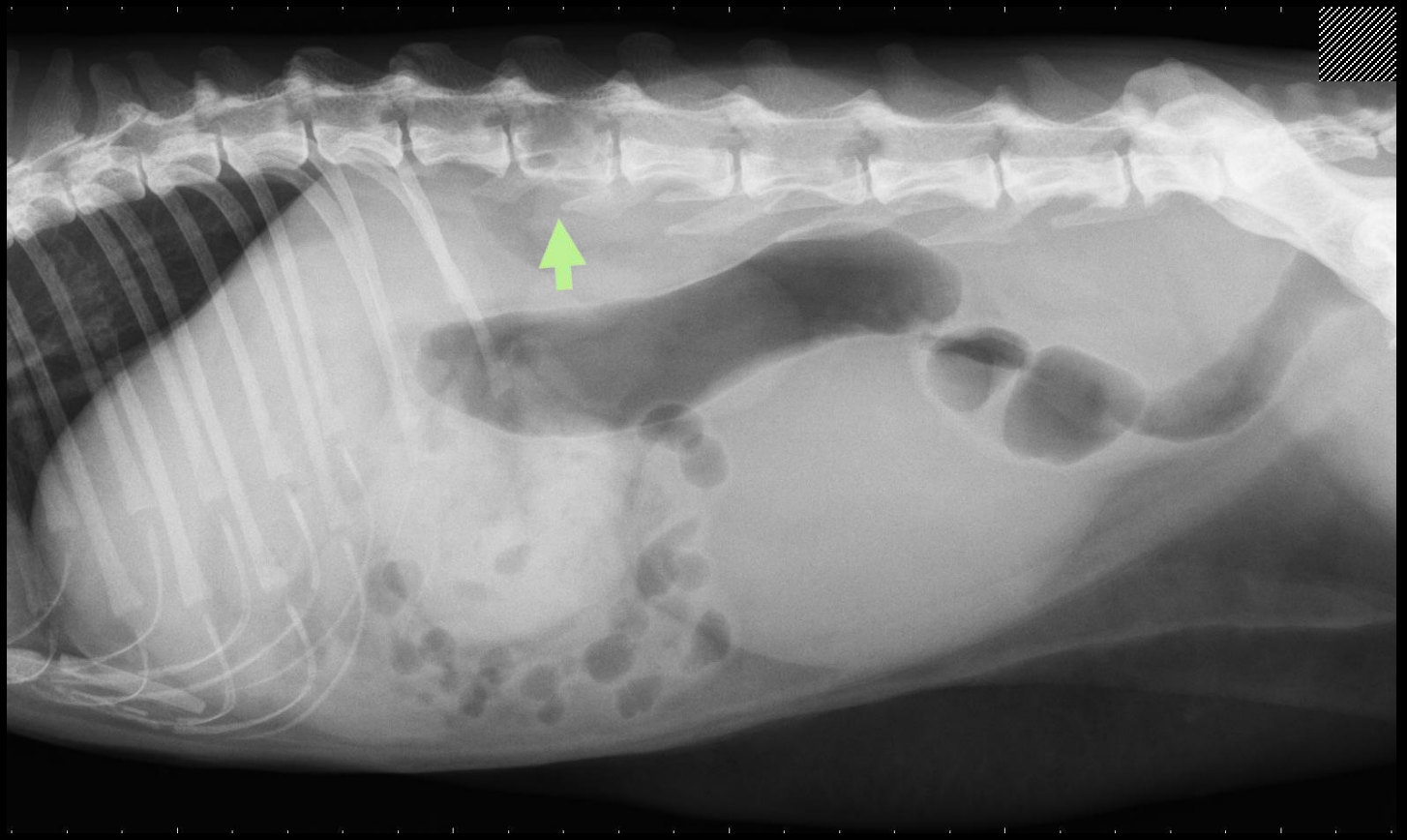
Neoplasia

Differential diagnosis

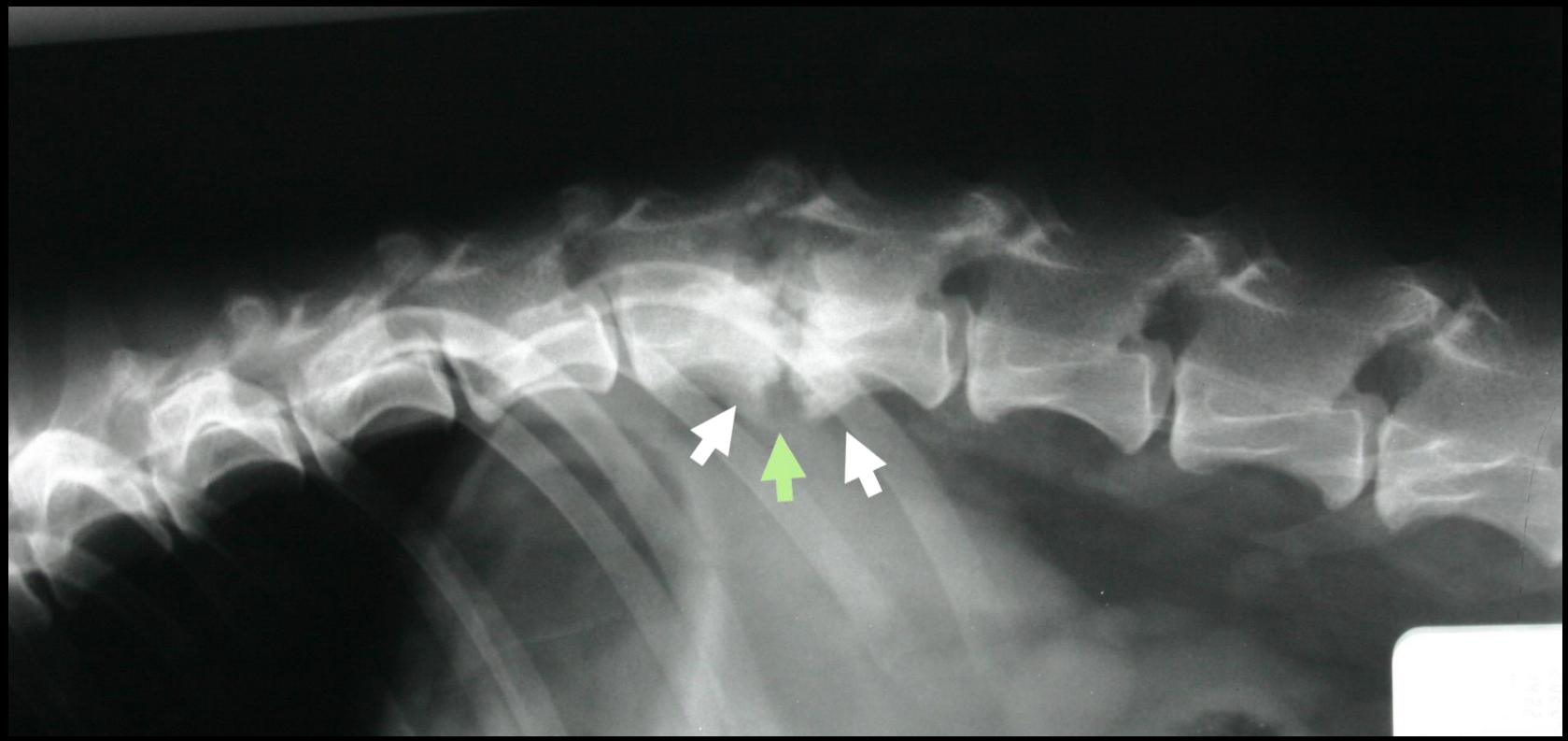
Opacity of the vertebrae

- Increased radiopacity
  - Osteopetrosis
  - Osteomyelitis
  - Neoplasia
  - Healing or healed fracture
- Increased radiolucency
  - Osteoporosis
  - Neoplasia
  - Osteomyelitis

Increased radiolucency: neoplasia



Increased radiopacity with increased radiolucency: osteomyelitis (in this case: discospondylitis)



Differential diagnosis

Size of intervertebral foramen

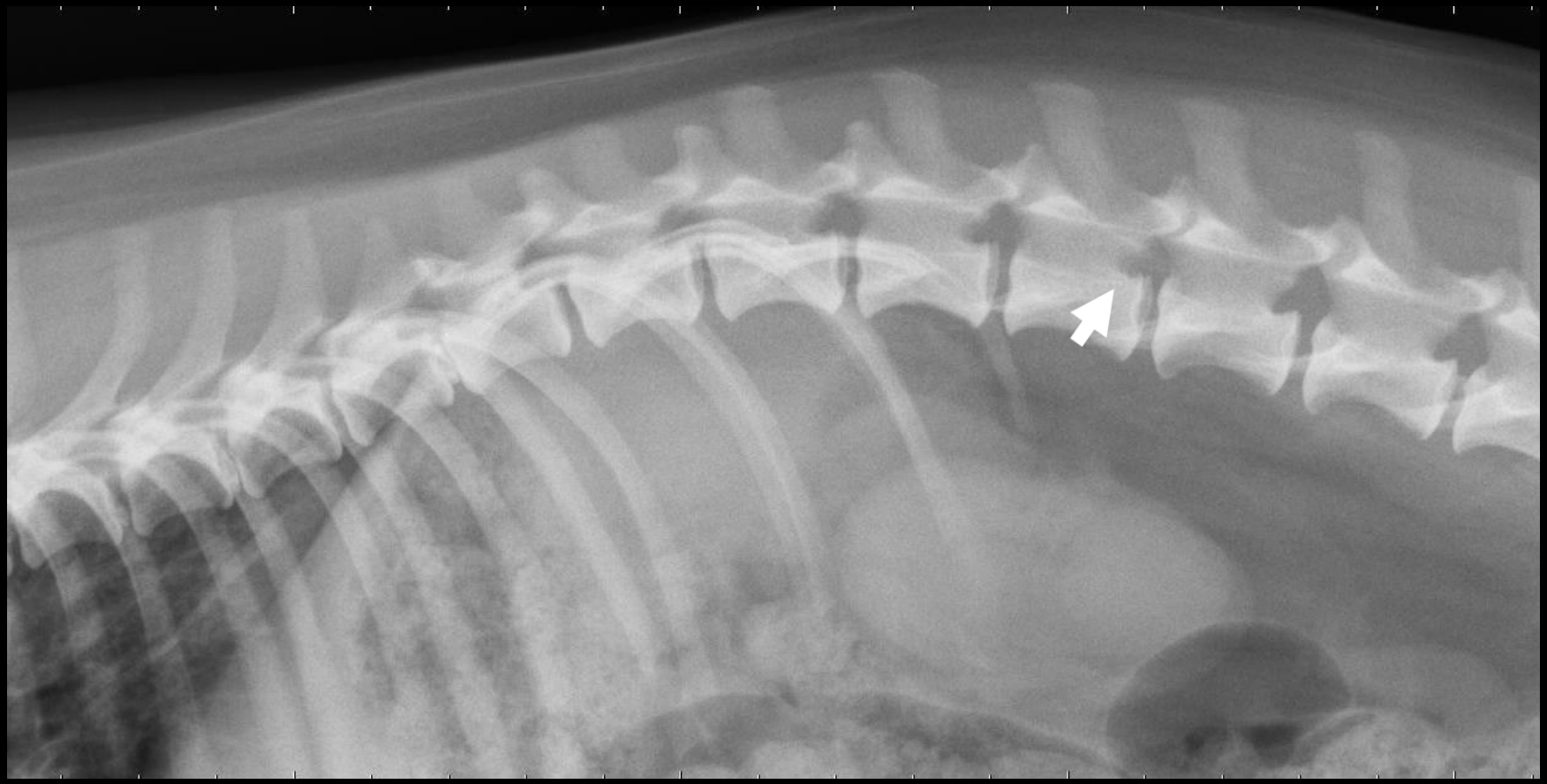
- Decreased foramen size
  - Disc prolapse with associated narrowing of the disc space
  - Bony proliferative tumor of adjacent vertebra
  - Trauma
  - Block vertebra
- Increased foramen size
  - Neoplasia of nerve root or surrounding soft tissue
  - After surgery
  - Trauma

Differential diagnosis

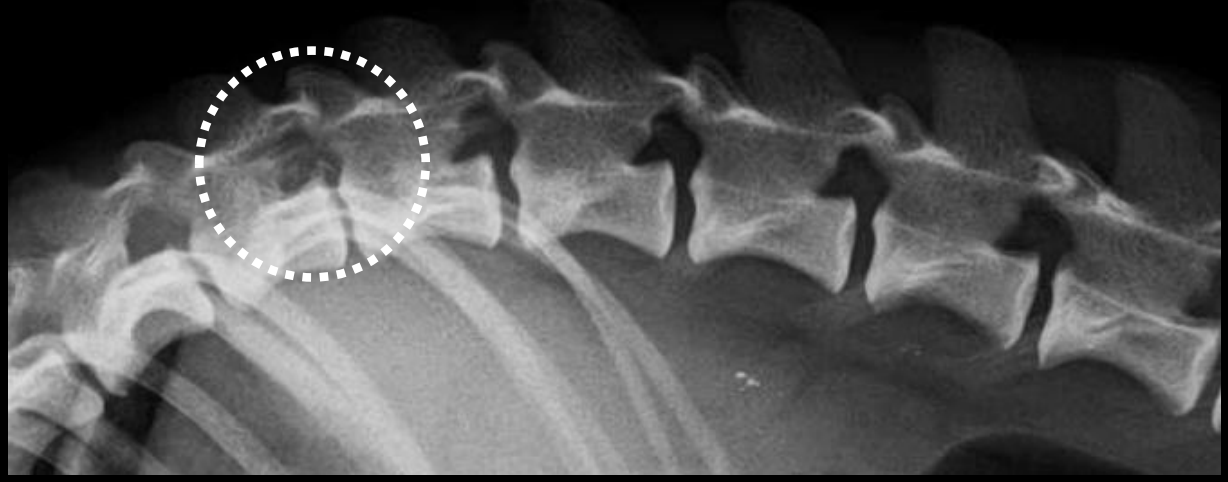
Increased opacity of intervertebral foramen

- Calcified disc material
- Ossifying pachymenigitis (dural ossification)
- Proliferative bony lesions of the adjacent vertebrae
- Foreign body

- Decreased size of intervertebral foramen: disc prolapse
- Increased opacity of intervertebral foramen: calcified disc material



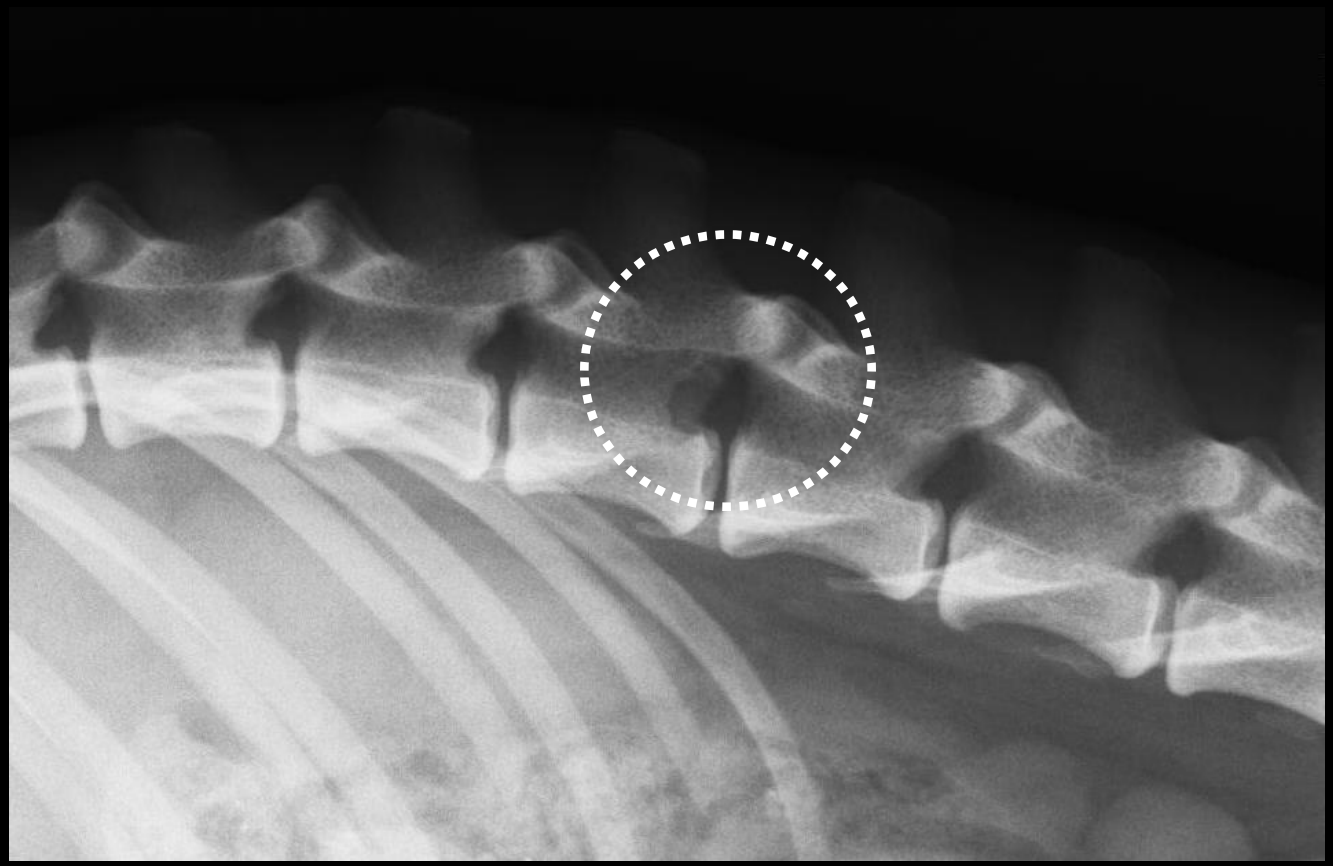
- Increased opacity of intervertebral foramen: calcified disc material

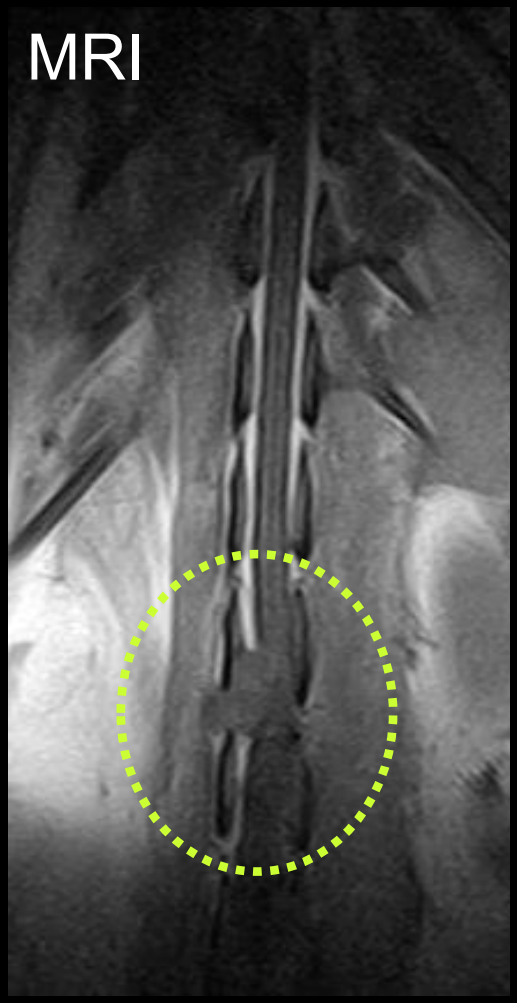


Myelography



- Increased size of intervertebral foramen: neoplasia





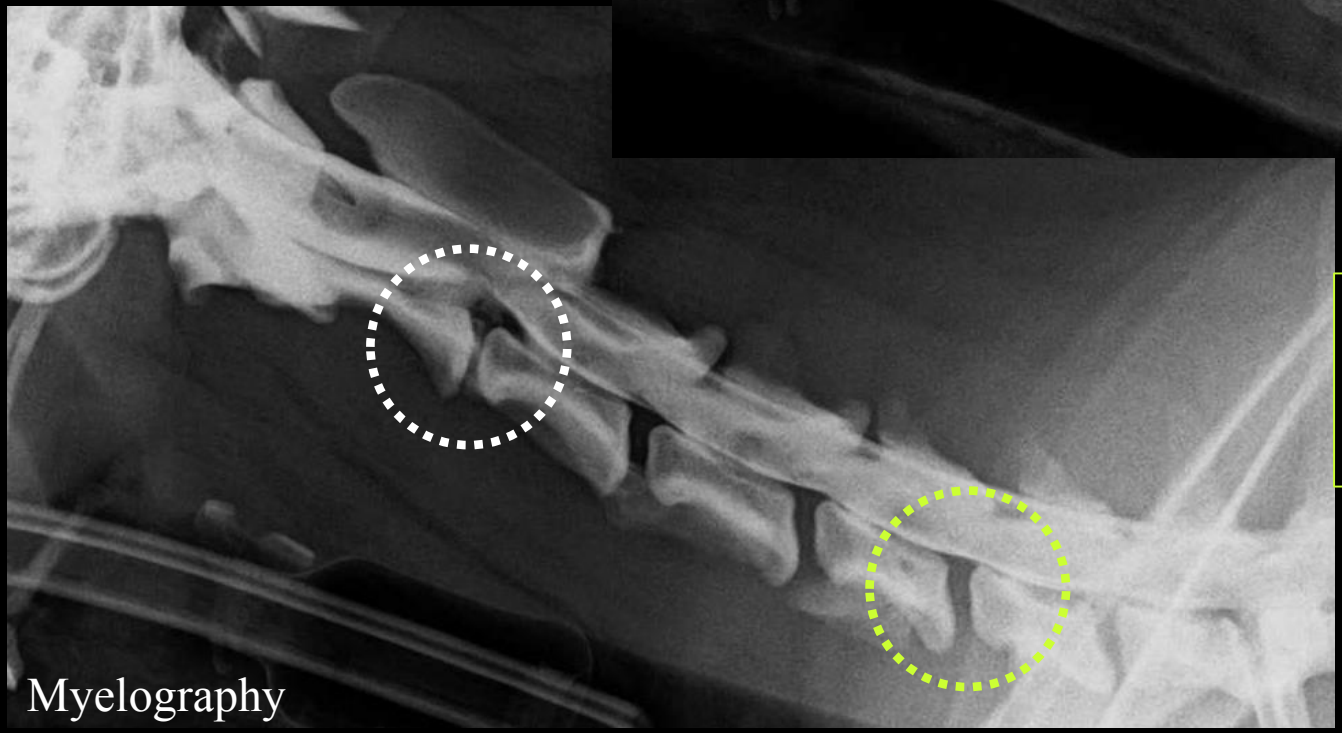
Differential diagnosis

Width of intervertebral disc space

- Decreased IVD width
  - Intervertebral disc extrusion or protrusion
  - Chronic discospondylitis
  - Fracture or luxation
  - Positional artifact
- Increased IVD width
  - Acute discospondylitis
  - Fracture or luxation

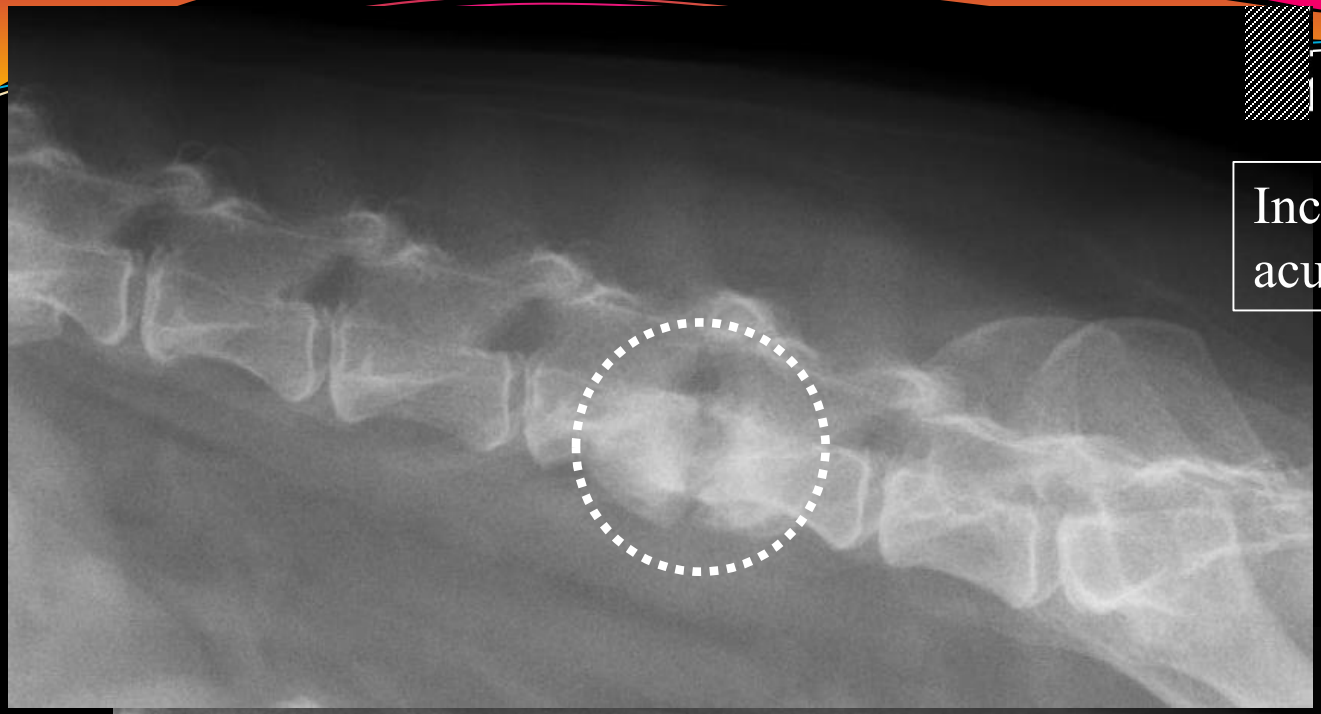
Decreased IVD width: intervertebral disc extrusion or protrusion



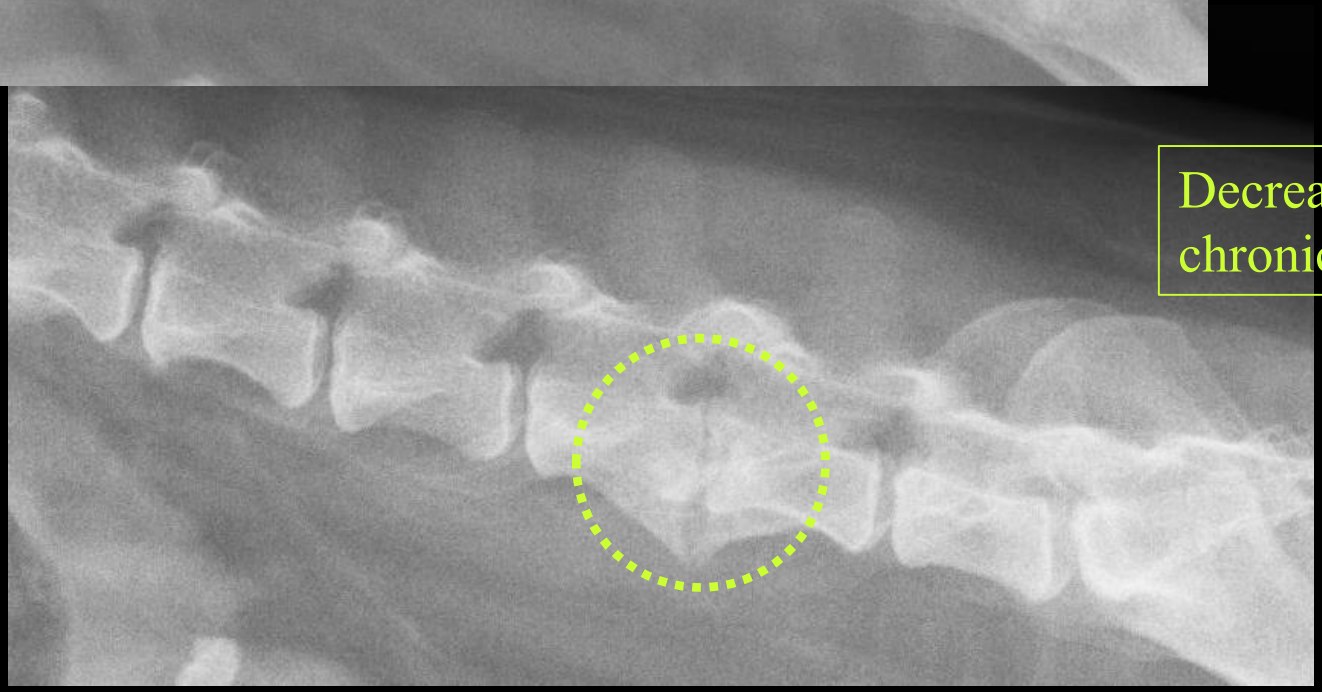


Myelography

Decreased IVD width in only one radiograph: poor position



Increased IVD width:
acute discospondylitis



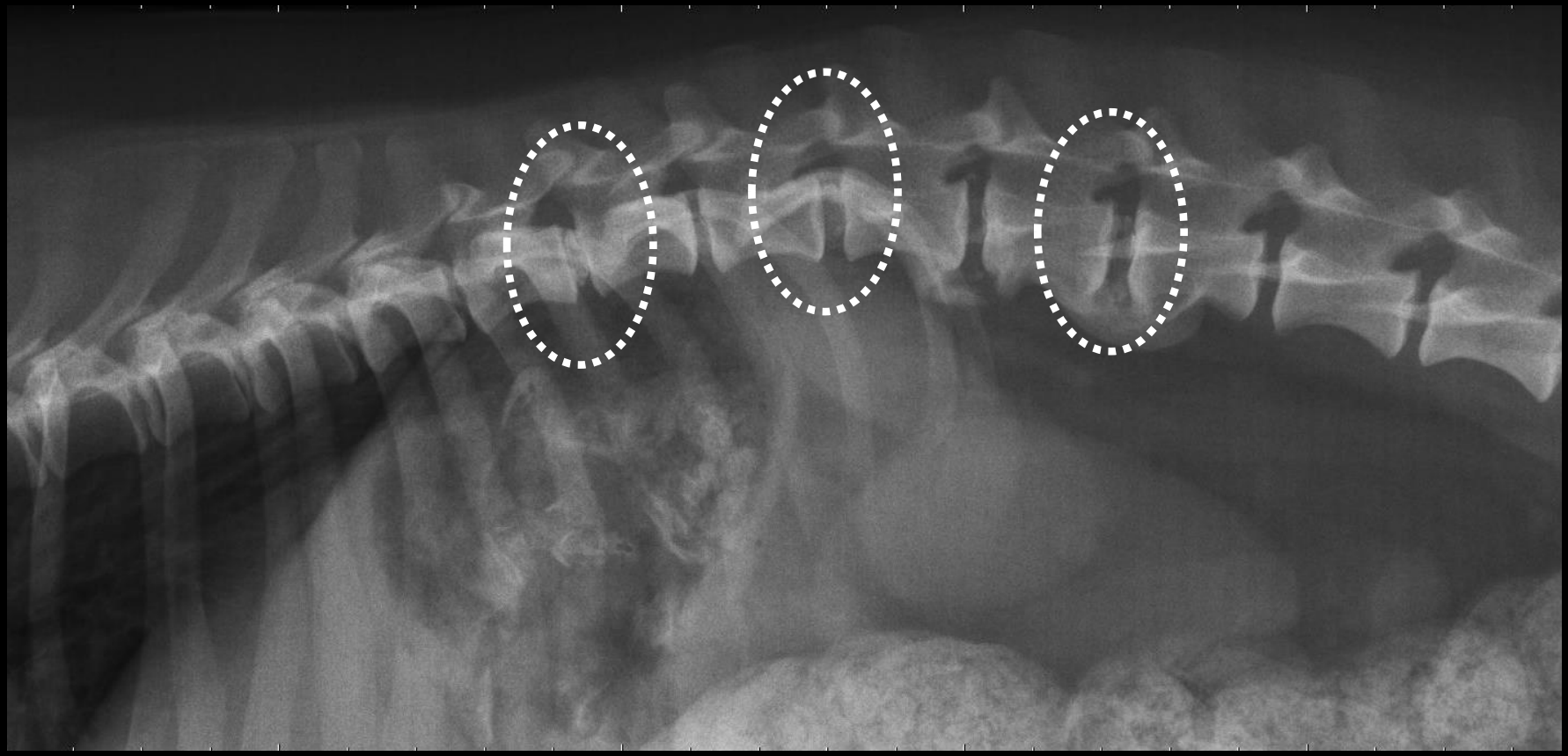
Decreased IVD width:
chronic discospondylitis

Differential diagnosis

Opacity of intervertebral disc space

- Increased radiopacity
  - Mineralized disc
  - Superimposed of lateral spondylosis deformans
- Increased radiolucency
  - Acute discospondylitis
  - Gas bubbles – vaccum phenomenon

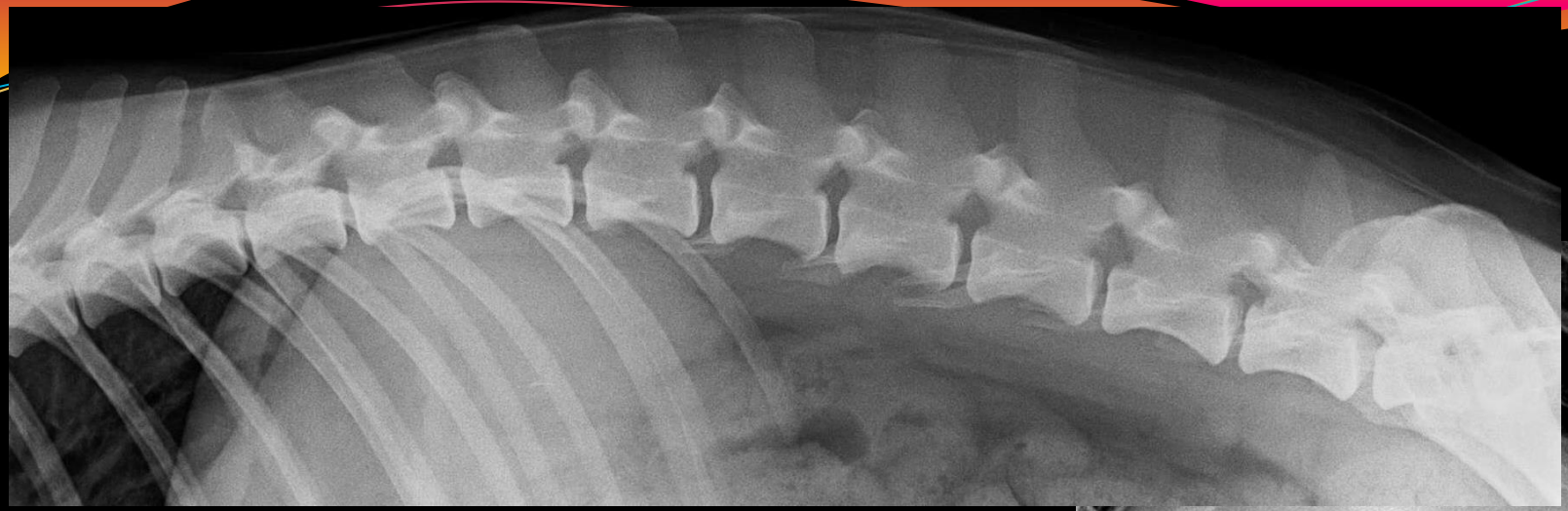
Increased radiopacity of intervertebral disc space

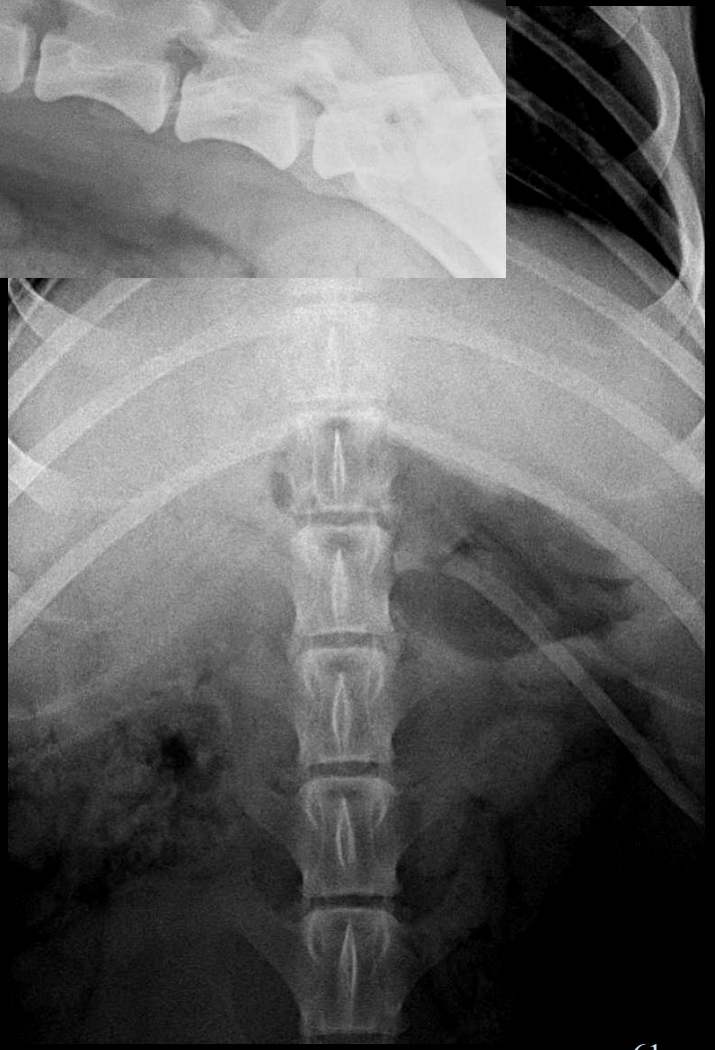


Specific diseases

Congenital vertebral anomalies

- Transitional vertebrae
- Block vertebra
- Hemivertebrae
- Butterfly vertebra
- Spina bifida





Transitional vertebrae

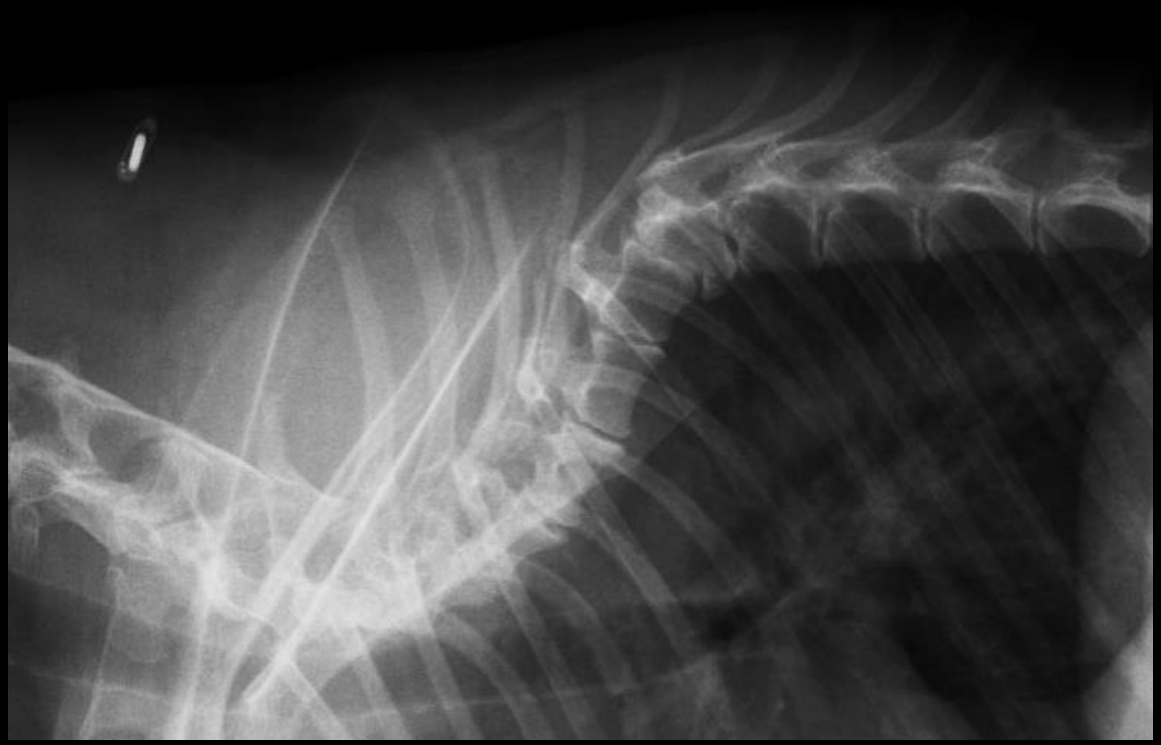
Block vertebra





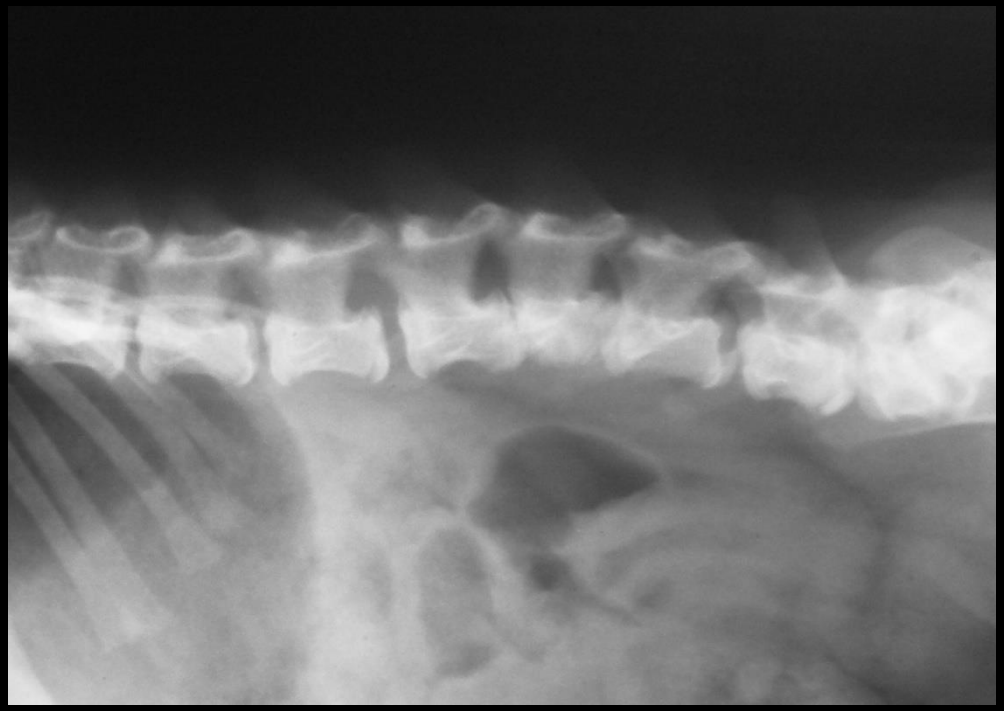


Hemivertebrae



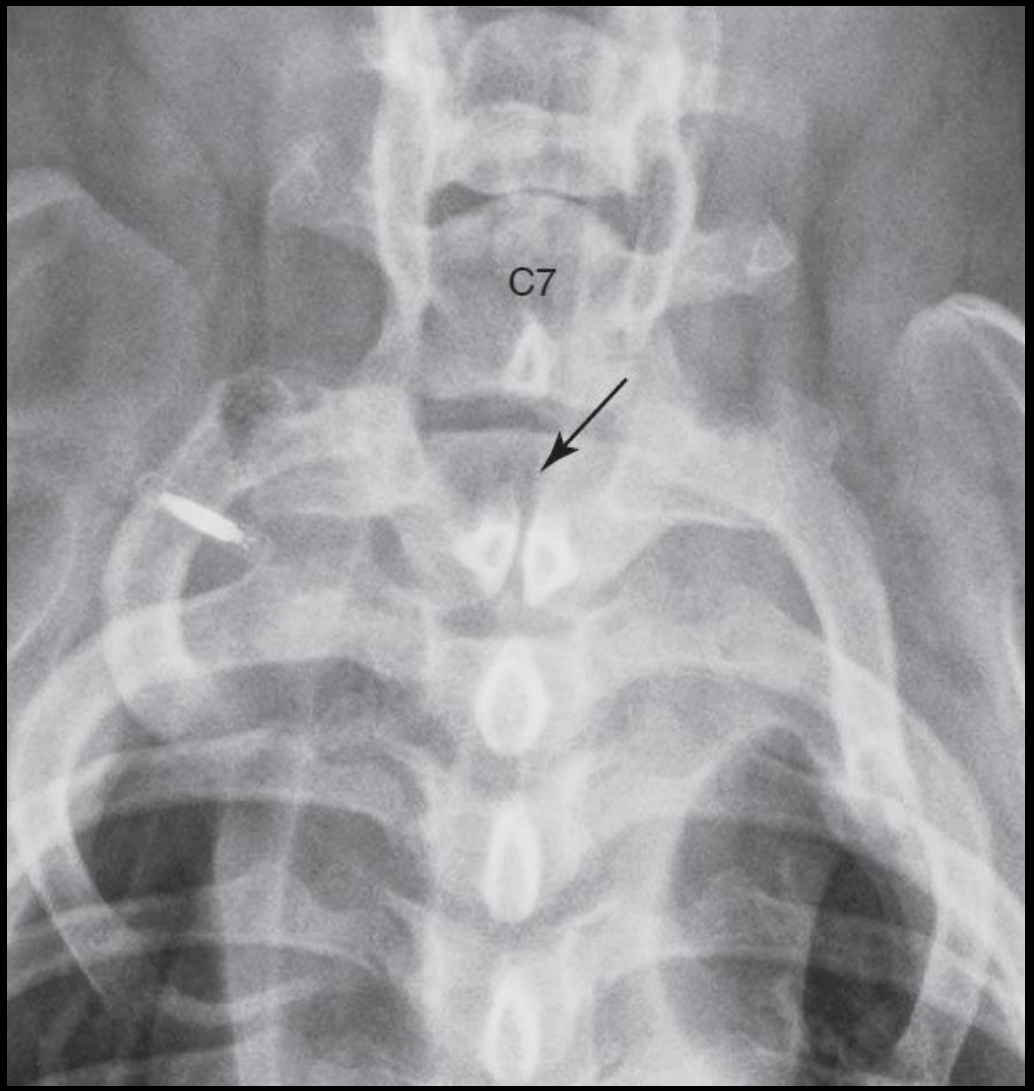


Butterfly vertebra



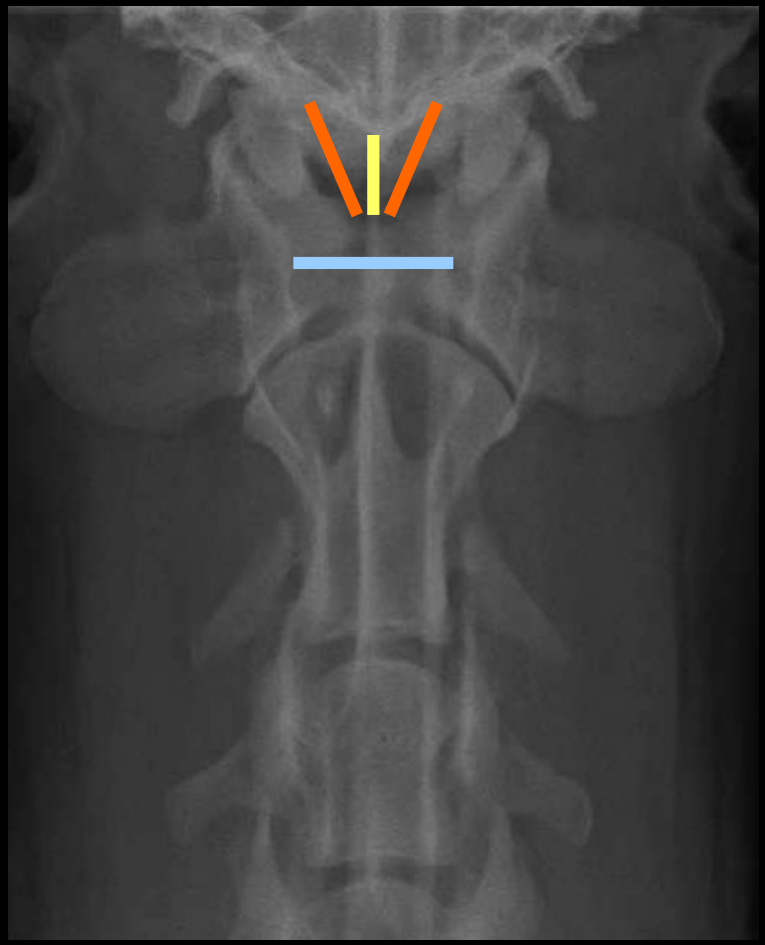


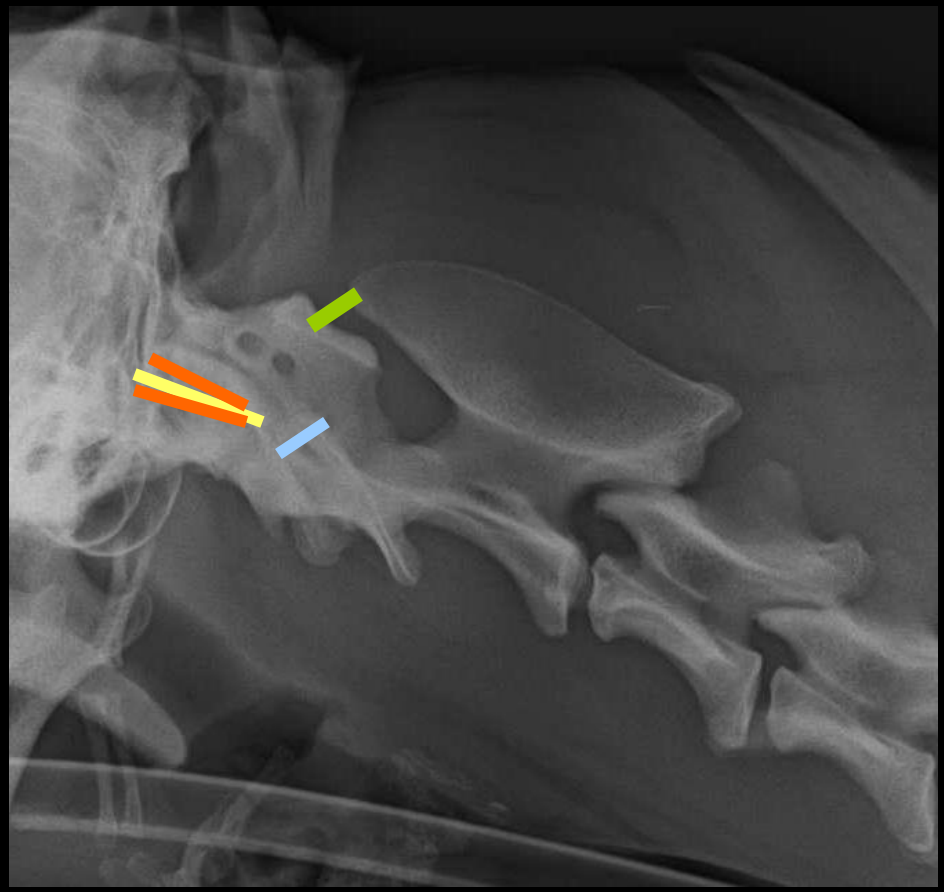
Spina bifida

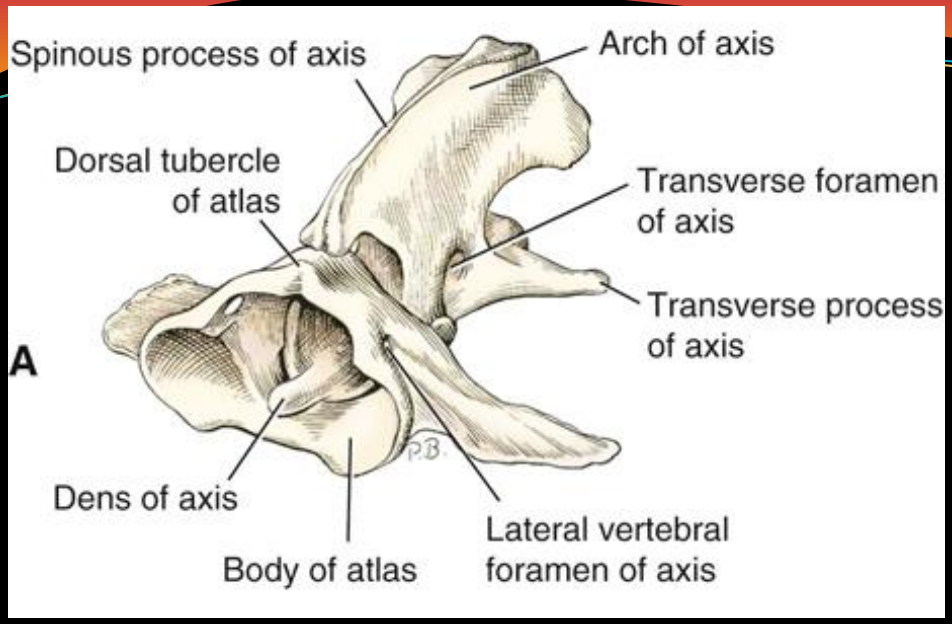


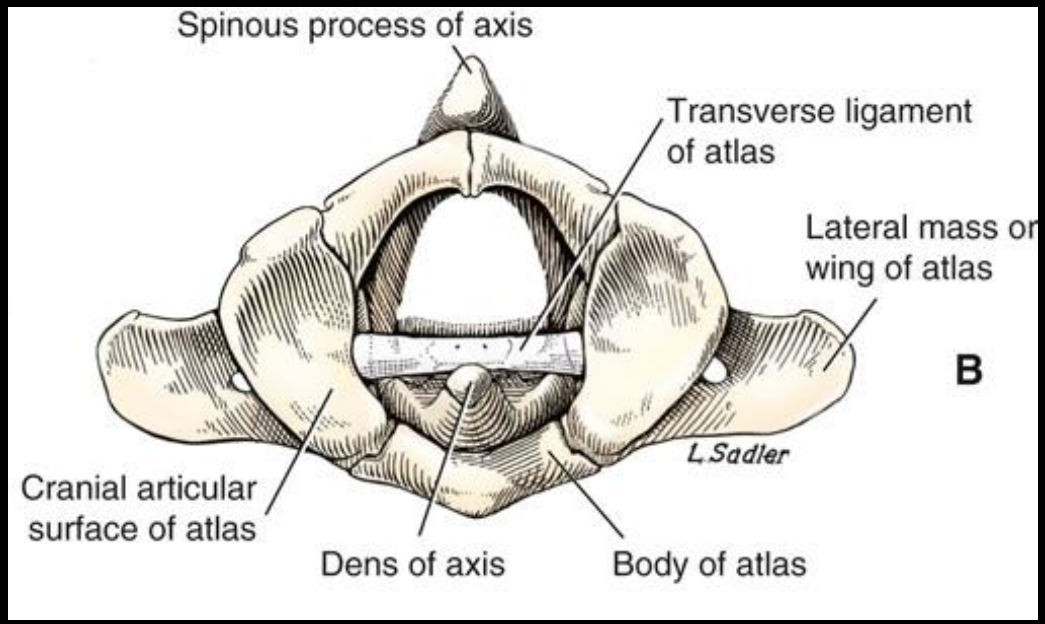
Textbook of Veterinary Diagnostic Radiology

Atlantoaxial instability / subluxation









Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, St Louis, 2010, WB Saunders.

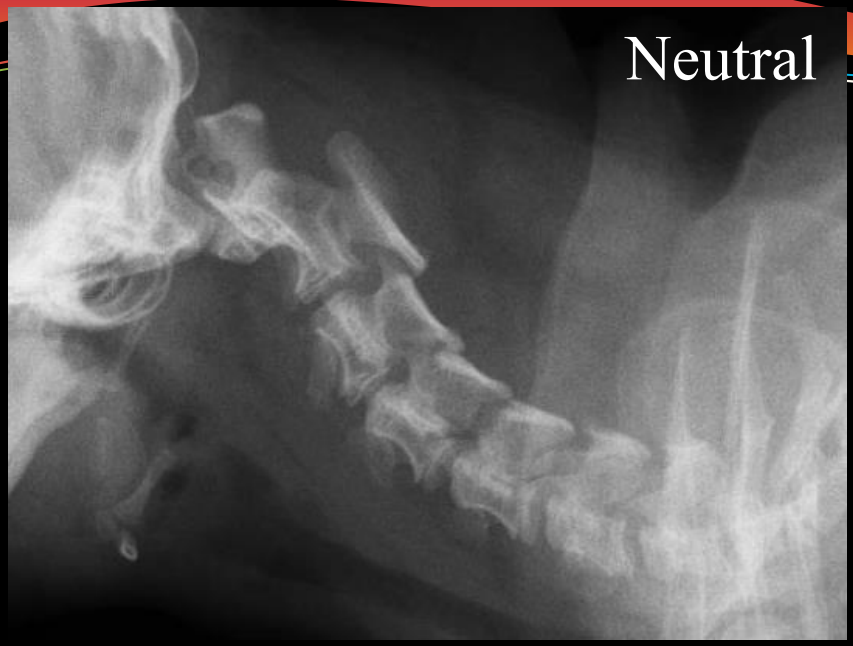
Case 1







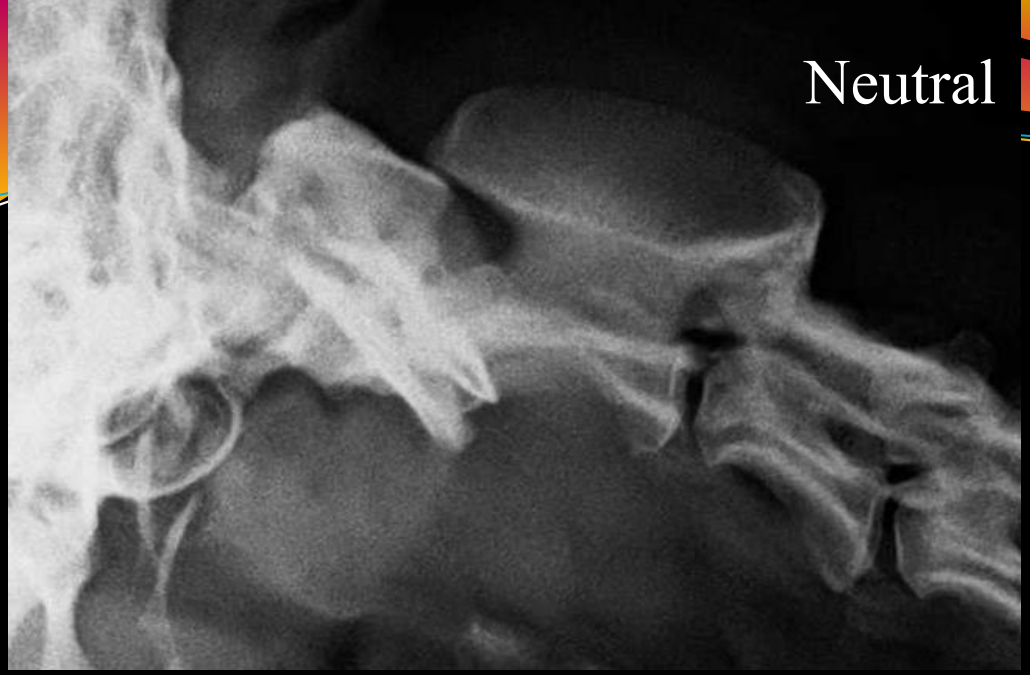
Case 2







Case 3









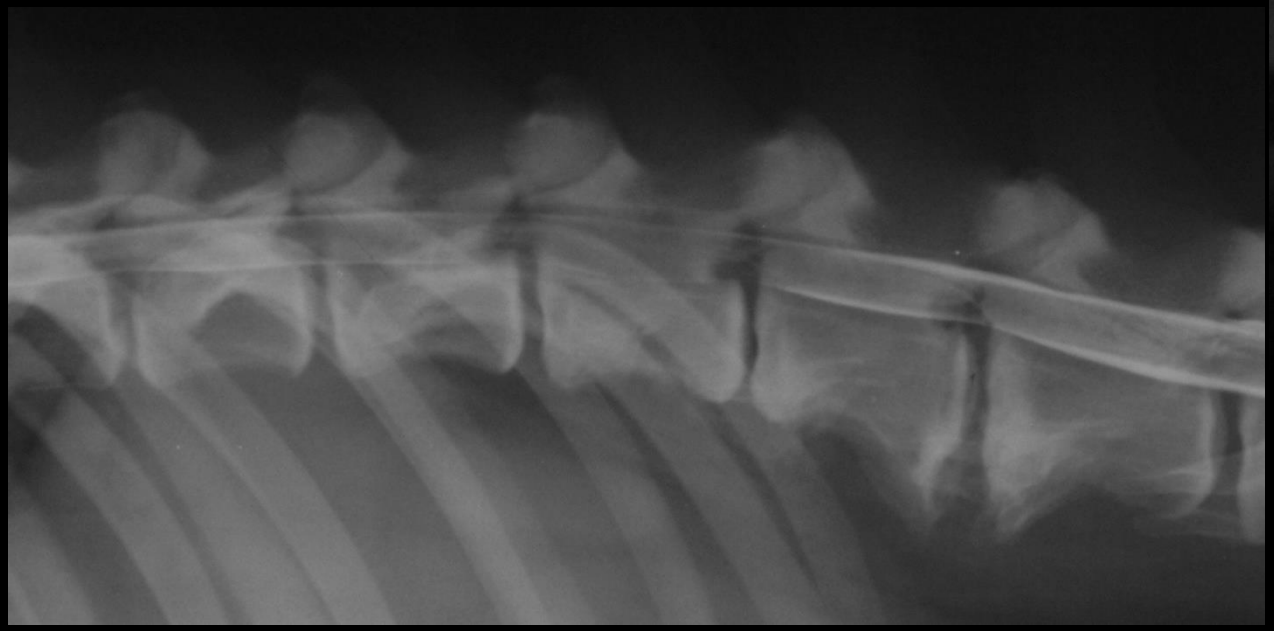
Spondylosis deformans

- Degenerative, proliferative disease of the vertebral column
  - Bony projections at intervertebral spaces (annulus fibrosis)
  - Osteophyte? Enthesophyte?
  - Small spurs to bony bridges across the disc space
- Subclinical disorder
- Stiffness, restricted motion, pain – in small % of patients
- More signs in sport or working dog











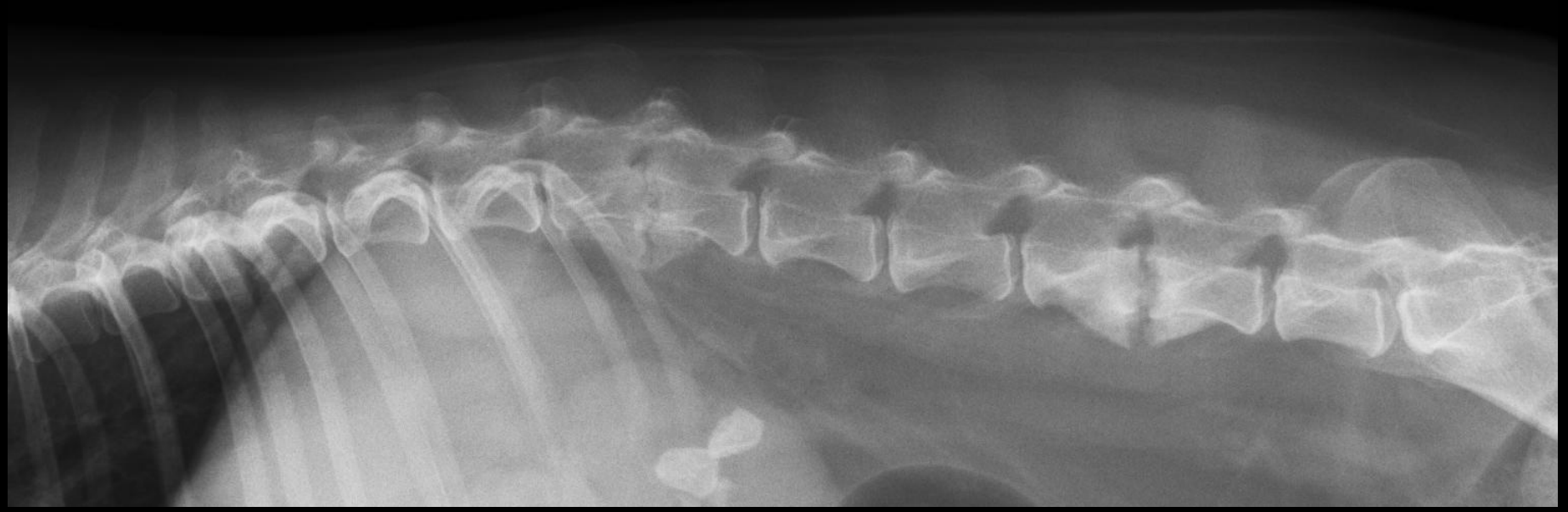
Discospondilytis

- Radiographic changes lag behind the onset of clinical signs
- More than one disc may be affected
- Early findings
  - Irregular lysis at endplates
  - Widening or collapse of IVD space
- Later findings
  - Collapse of IVD space
  - Sclerosis of the vertebral endplates
  - Enthesophyte formation at the margins of the vertebral bodies
  - Fusion of adjacent vertebral bodies may occur with healing

Case 1 – 1st visit



Case 1 – 4th week



Case 1 – 10th week



Neoplasia

- Source
  - Vertebra
  - Extravertebral soft tissue
- Location
  - Involving the vertebral canal
  - Outside the vertebral canal

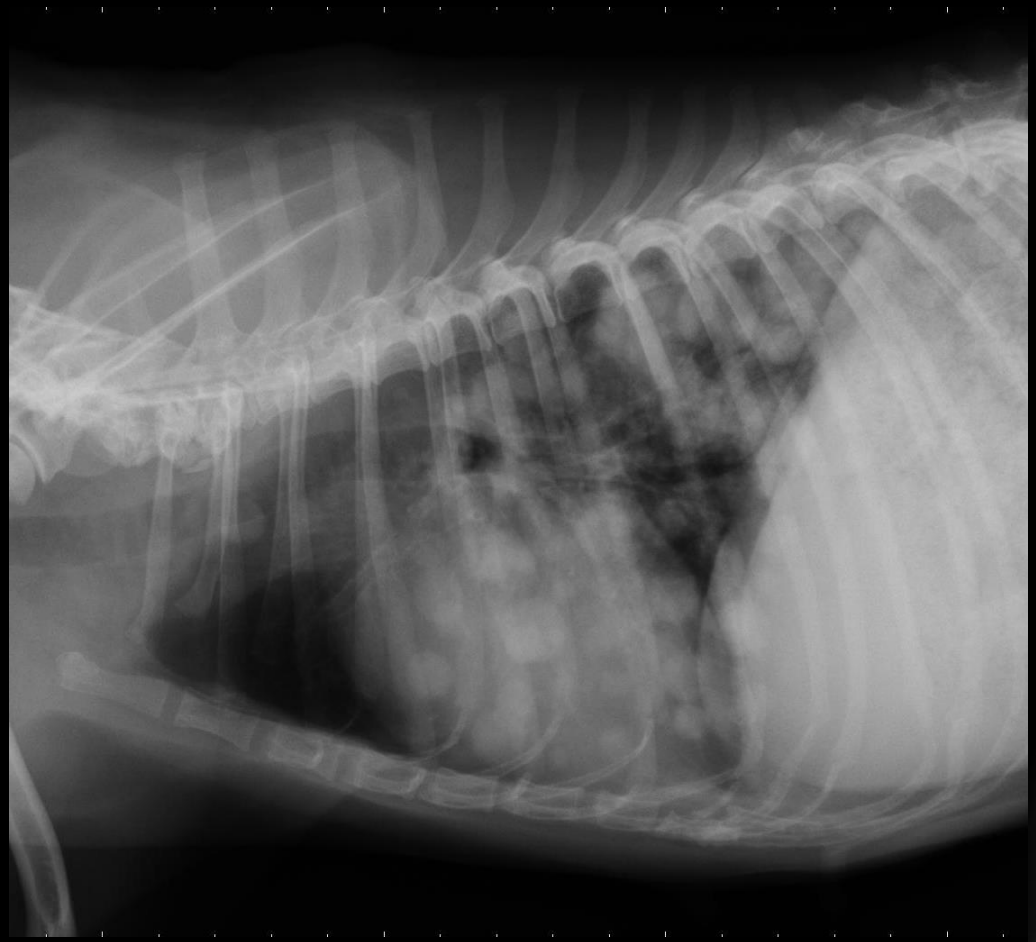
Vertebral neoplasia

- Benign tumors of vertebra
- Malignant tumors of vertebra
  - Primary:
    - One vertebra
    - Aggressive with osteolytic, osteoblastic or mixed changes
  - Metastatic:
    - > one vertebrae +/- other bones
    - Osteolytic
    - Metastatic carcinoma: productive bony lesion involving the caudal lumbar and sacral vertebrae, the pelvis, adjacent vertebrae, and the femurs
  - +/- Pathological fracture

Case 1







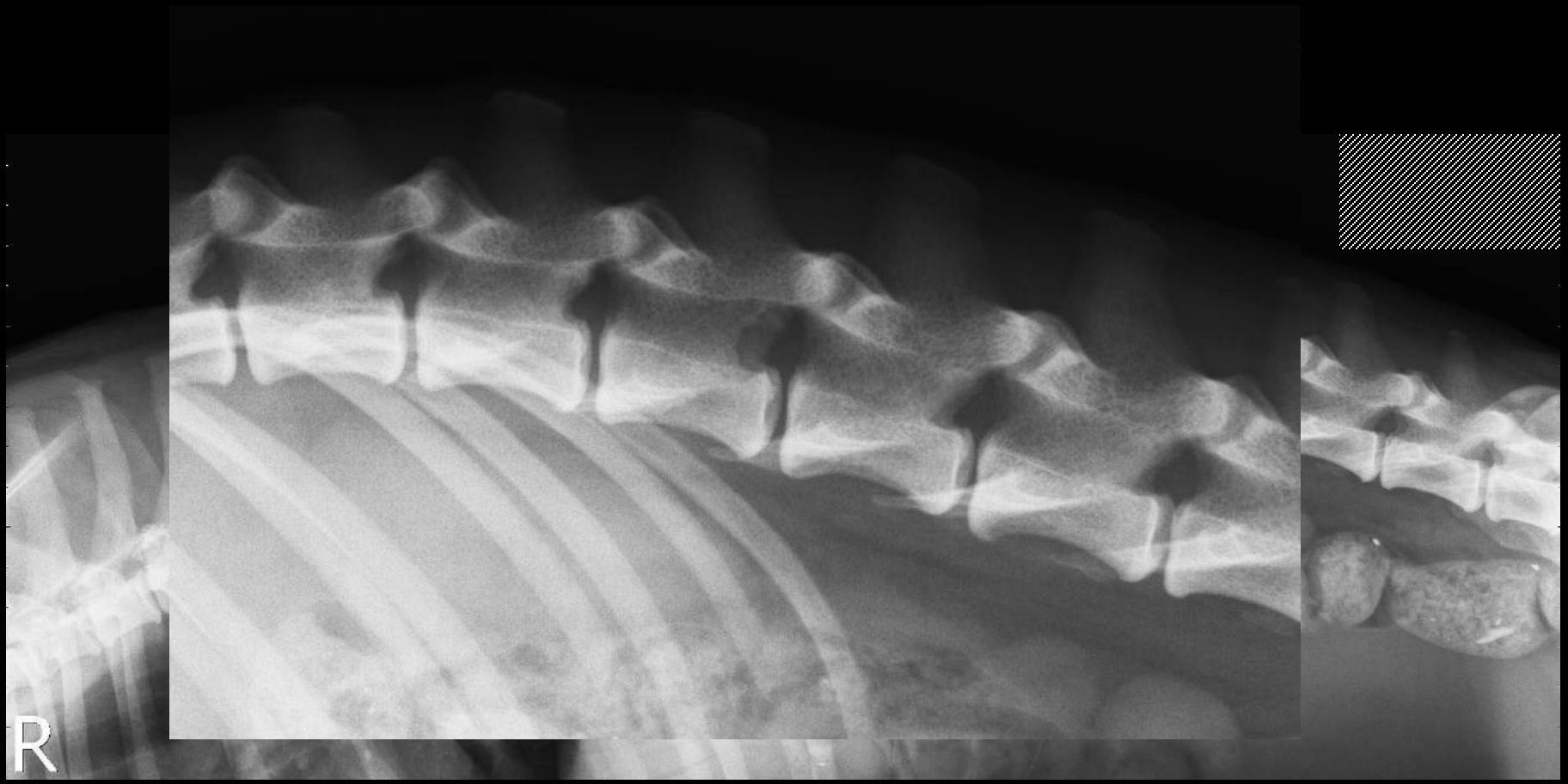


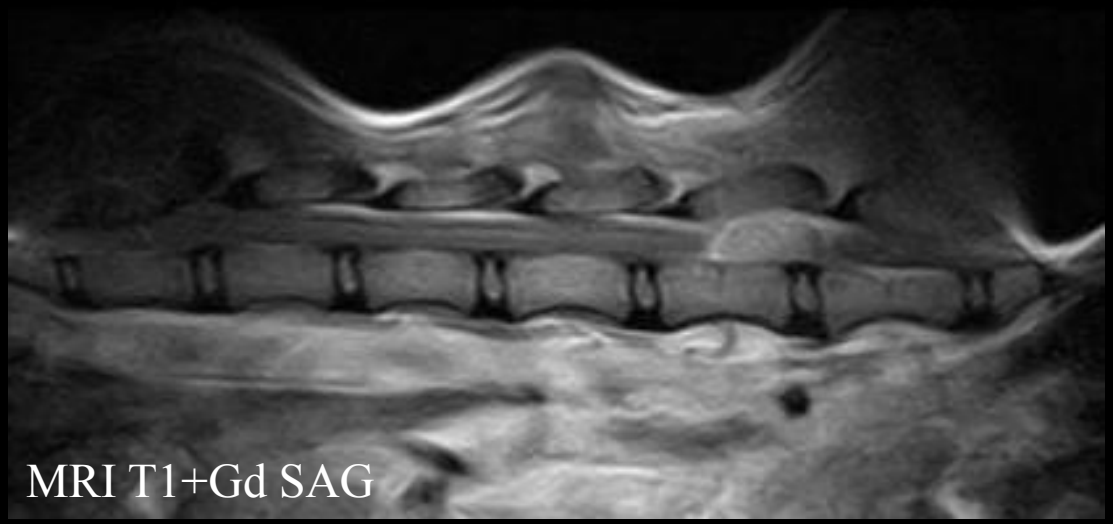
Case 2

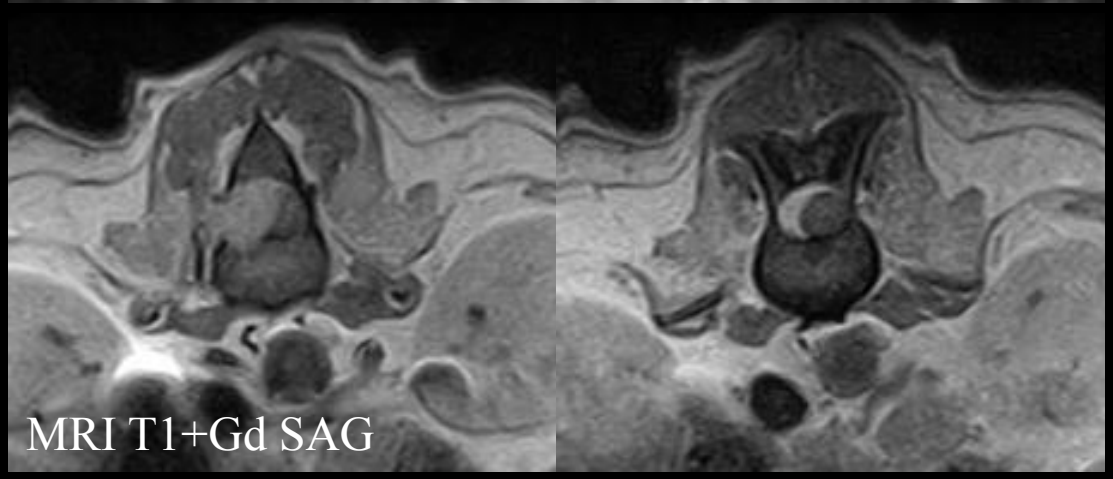




Case 3




MRI T1+Gd SAG

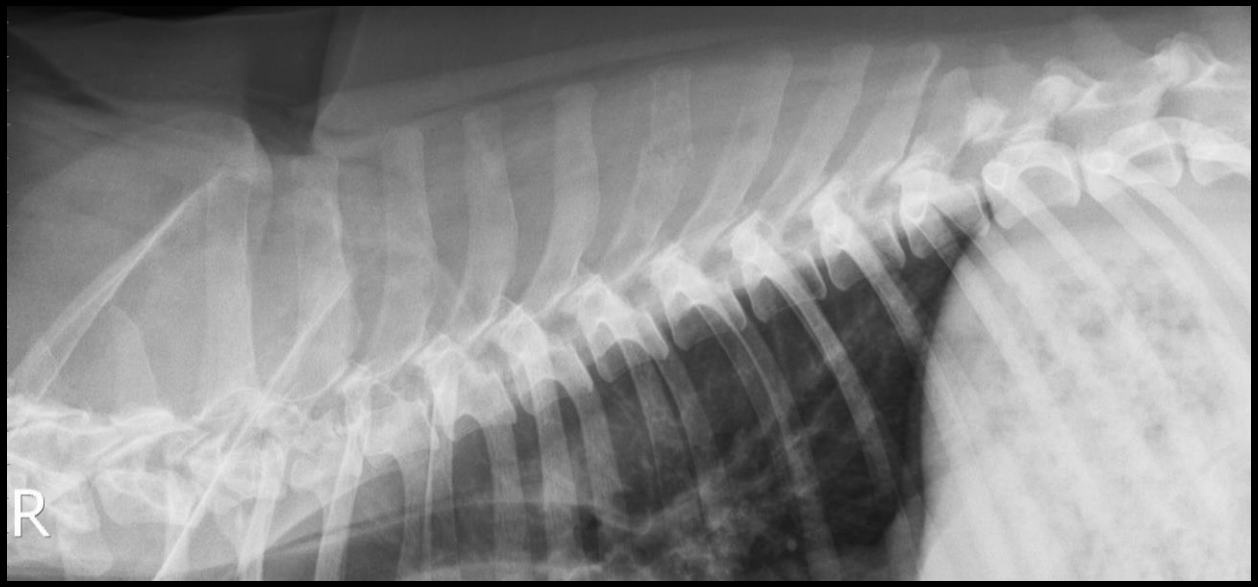

MRI T1+Gd SAG

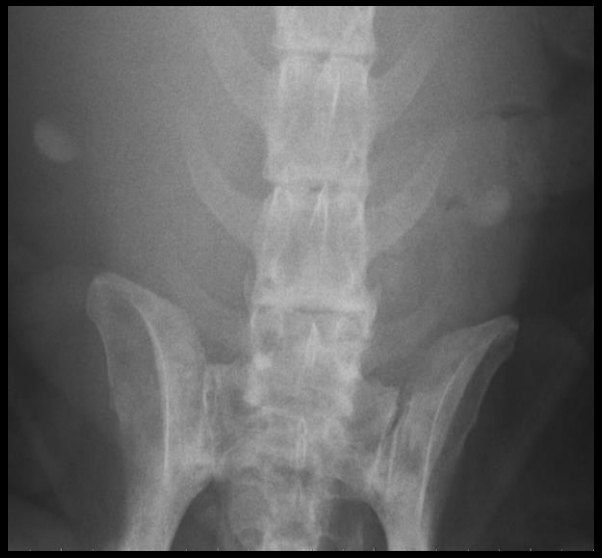

MRI T1 DOR

Case 4



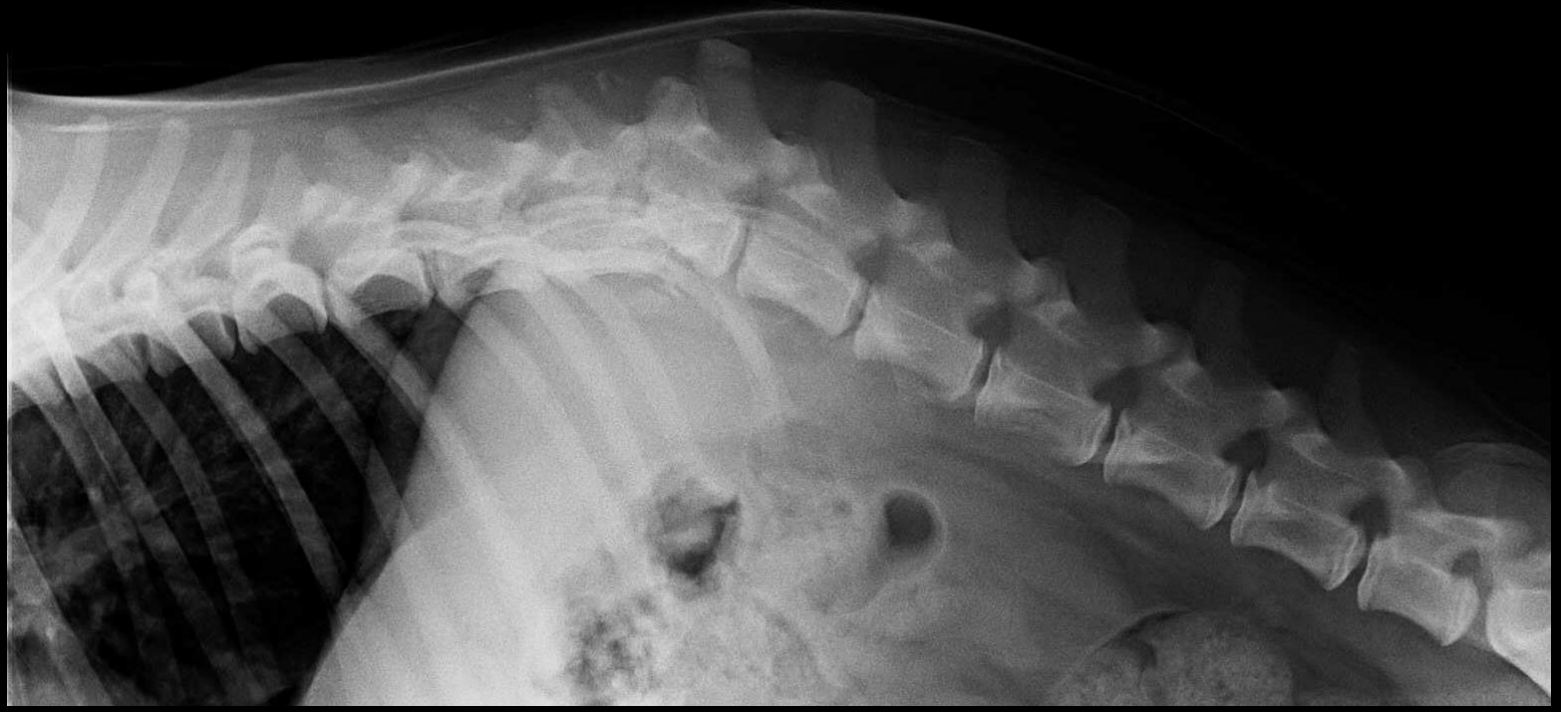




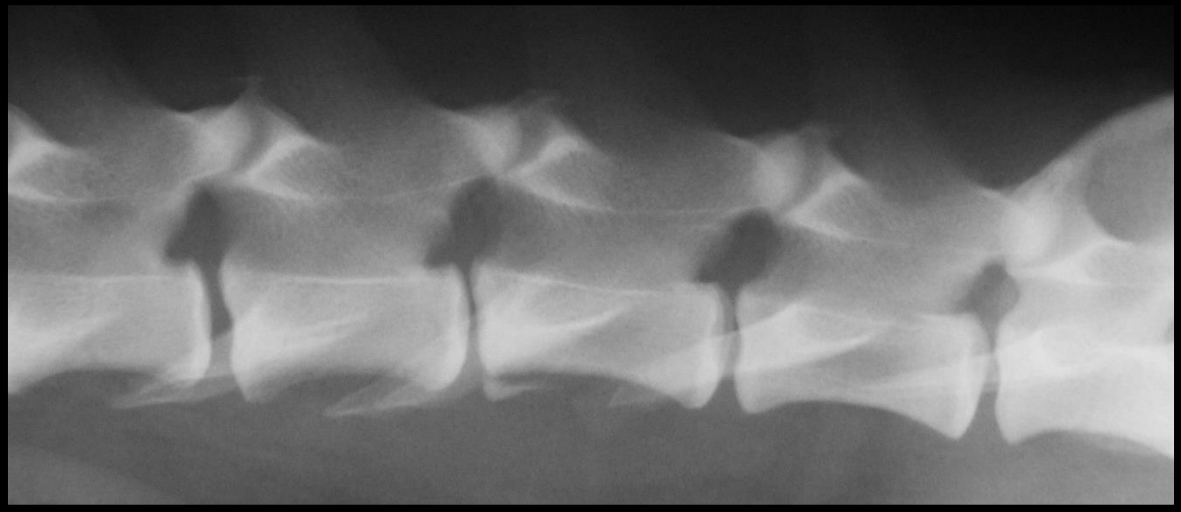


Trauma

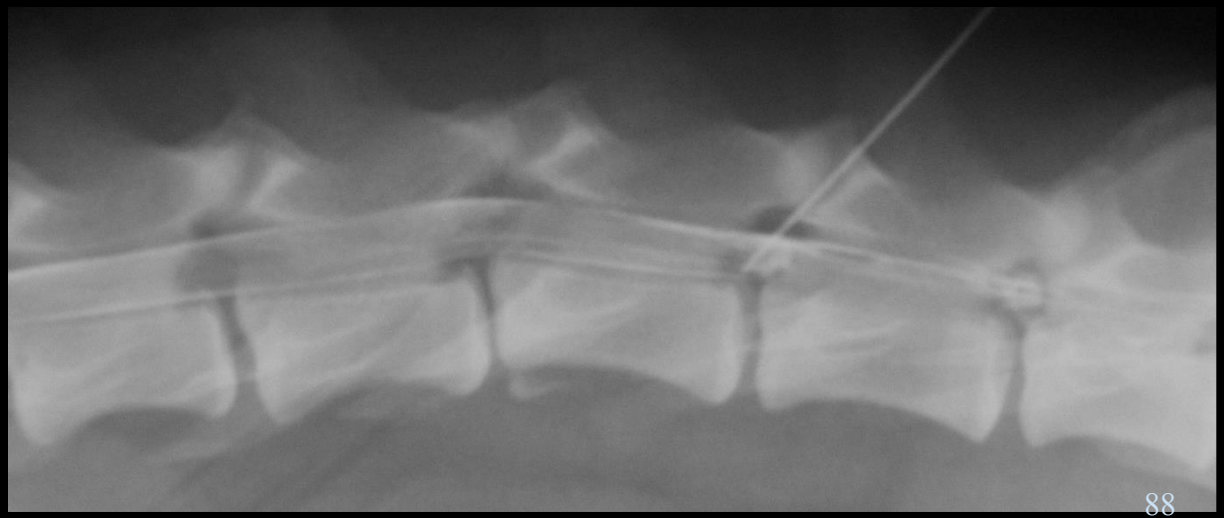
- Fracture
- Subluxation / luxation



L1 fracture and T13/L1 subluxation

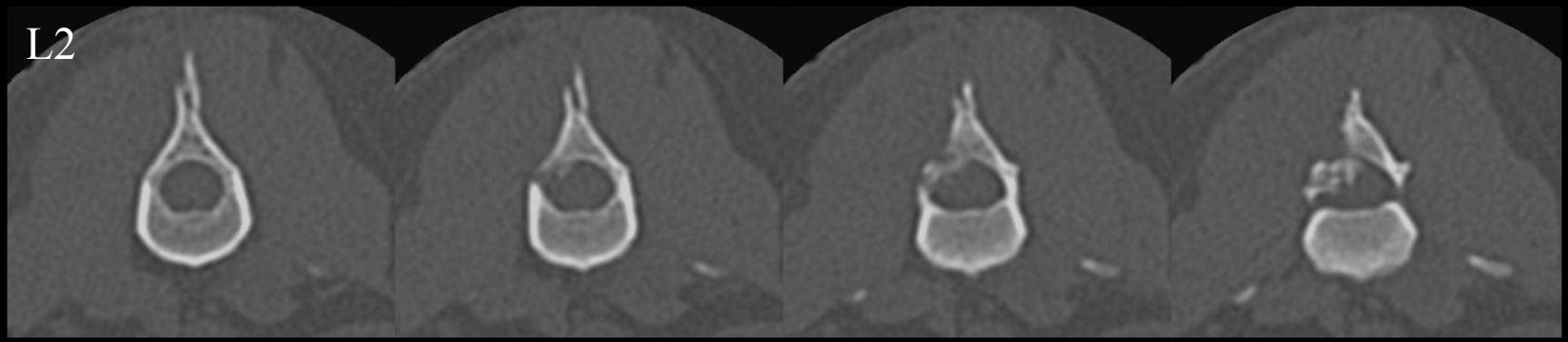


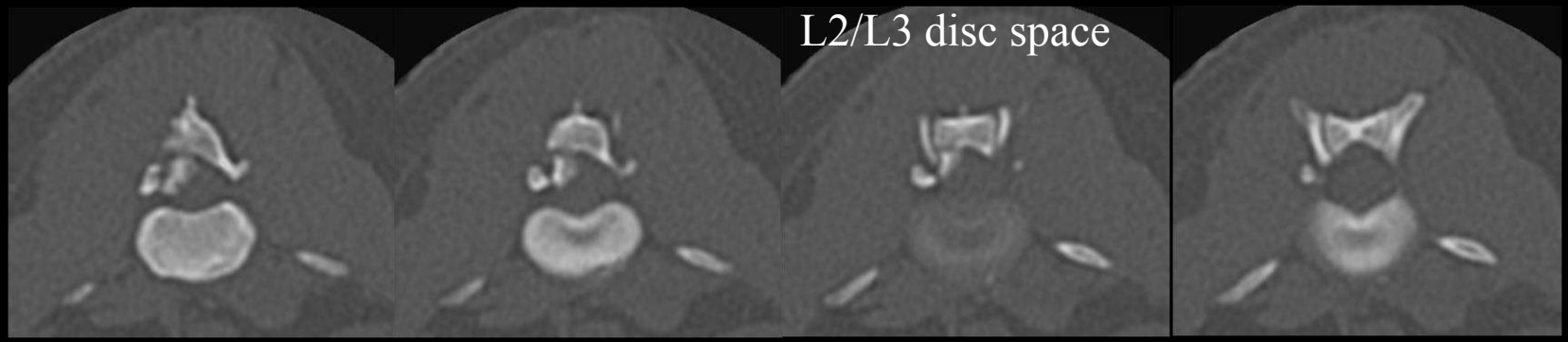
L4-5 subluxation



CT – spinal fracture



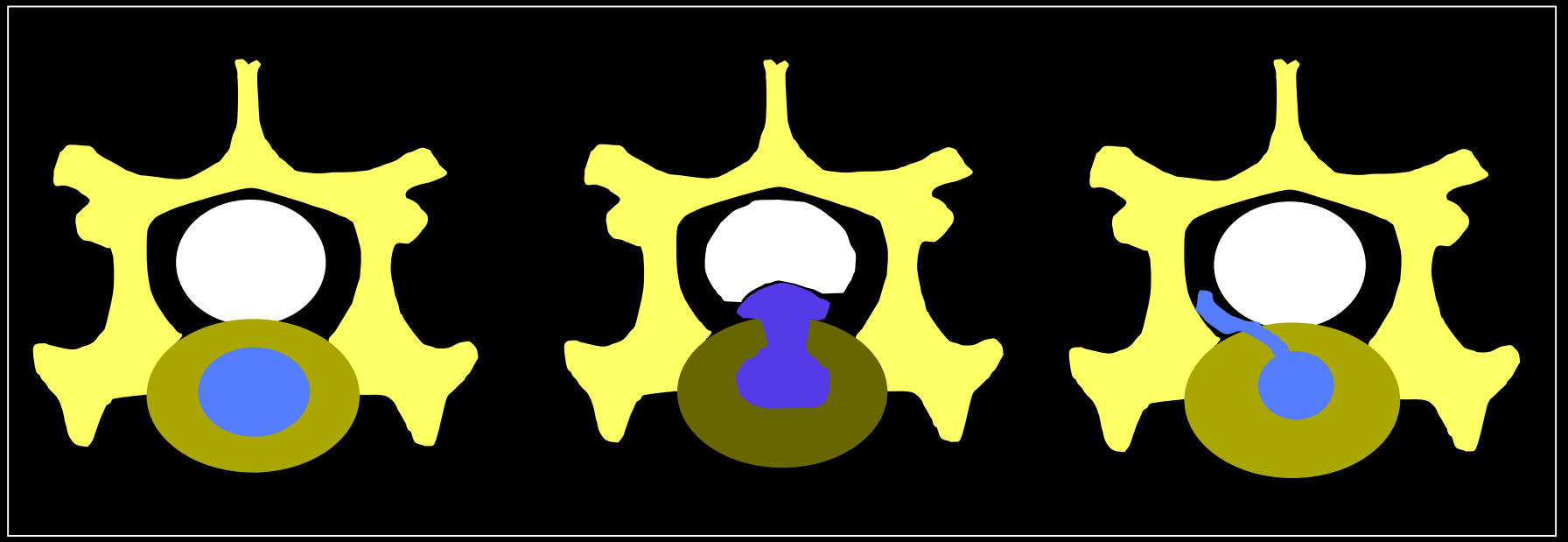




Non-degenerative

Normal disc

Traumatic disc extrusion



Degenerative disc disease

Degenerative intervertebral disc disease

- Radiography
  - Narrowed disc space
  - Narrowed intervertebral foramen
  - Calcified disc material
    - In disc space
    - In intervertebral foramen

Degenerative IVDD

- Myelography
  - Extradural compression
- CT
  - Calcified materials in the vertebral canal
- MRI
  - Degenerative disc
  - Spinal compression
  - +/- spinal edema (not common)













CT – degenerative disc extrusion

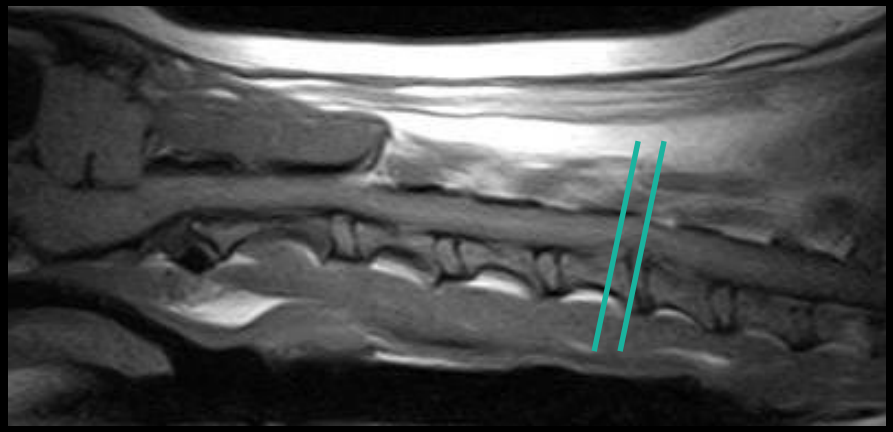




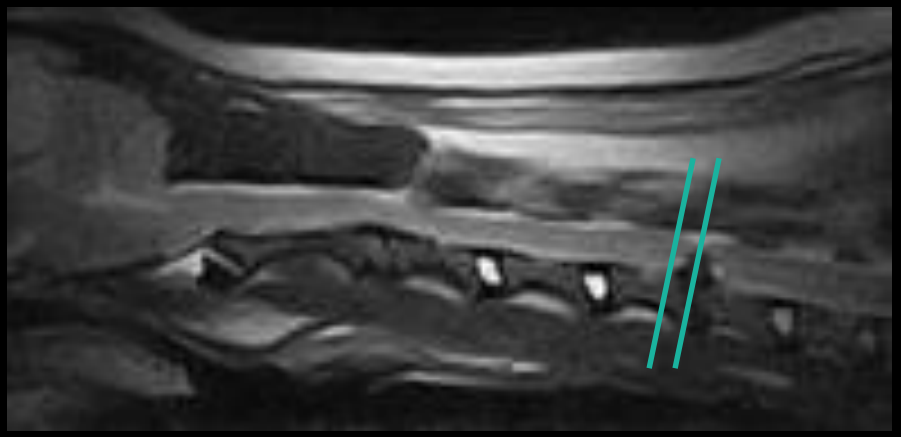


MRI – degenerative disc extrusion

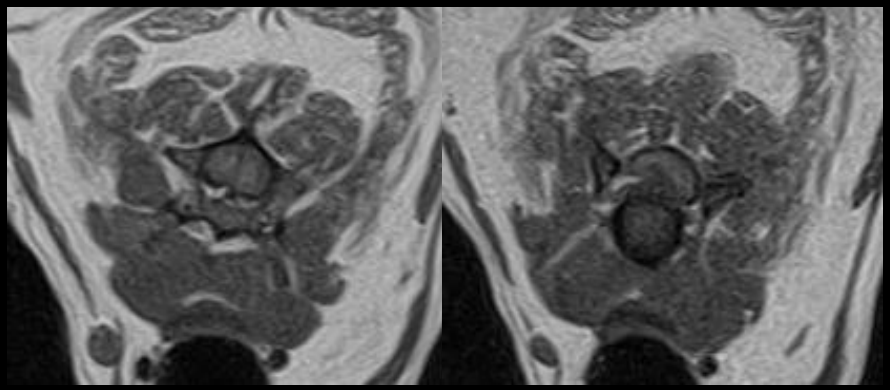
T1 SAG



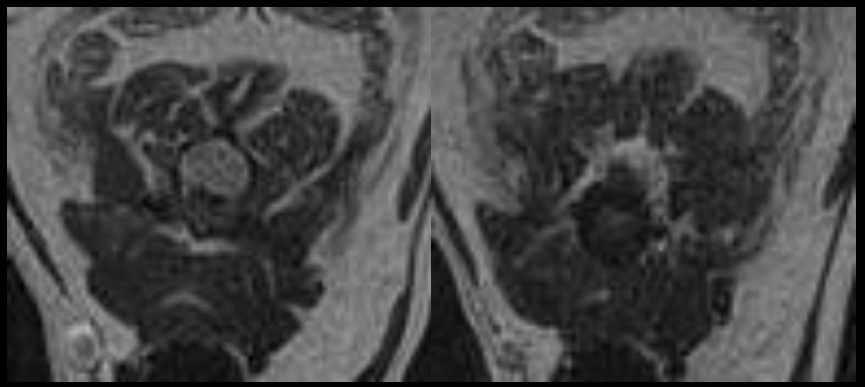
T2 SAG



T1 TRA



T2 TRA



Acute non-compressive nucleus pulposus extrusion (ANNPE)

- Different from typical degenerative disc disease
- Associated with trauma
- Mainly spinal contusion
  minimal spinal compression

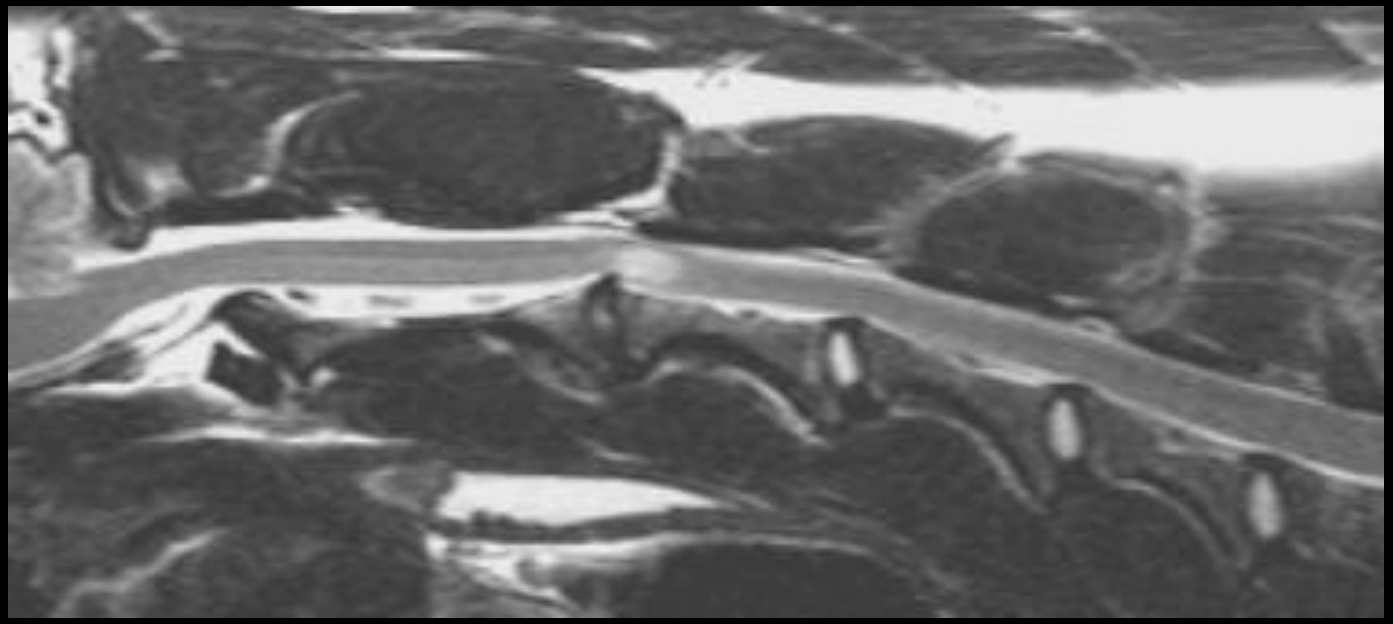
Clinical signs

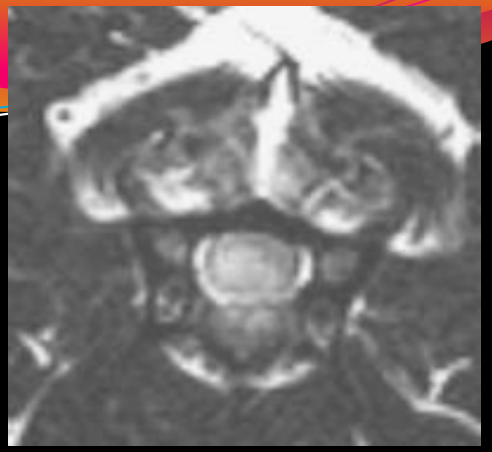
- Acute onset, non-progressive
- Neuro deficits with focal spinal cord lesion
- +/- spinal pain

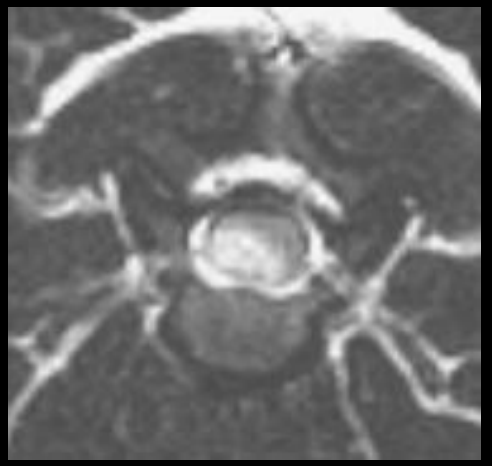
Acute non-compressive nucleus pulposus extrusion (ANNPE)

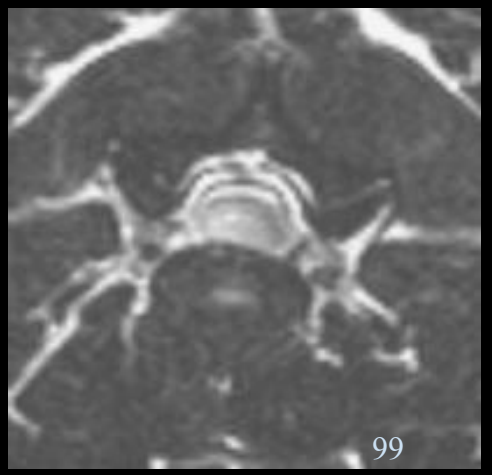
- Radiography
  - Narrowed IVD space
  - No spinal instability
- Myelography
  - Focal intramedullary lesion near IVD space
  - No or minimal extradural compression
- MRI
  - Narrowed disc space
  - Normal hydration in IVD, reduced IVD volume
  - Hyperintensity on T2W near IVD space
  - +/- Intramedullary hemorrhage











- Radiography
  - Get the most of it !!!
  - Systemic interpretation
  - Limitation

大五診療實習影像診斷X-RAY報告

組別：　　學號：　　姓名：　　日期：

Animal Information	
Reg. No.	Species:
Pet Name:	Breed:
Sex: M F	Neutered? Y N
Age:	Body weight:
Examination Date:	
Reasons for Examination	

Imaging Findings: (1. Image quality evaluation, 2. Findings description)

Radiological Differential Diagnosis and Recommendation:

Discussion & Reference



- 4 year-old male German Shepherd Dog
- 3-week progressive history
  - Reluctant to jump, yelping when sitting down, and abnormal gait
- Clinical examination
  - Hyperesthesia at lumbosacral area
- Neurological examination
  - Suggestive of L6-S1 spinal cord segments or nerve roots problem



- Imaging findings
  - Enthesophytes in various degree at L3/L4, L4/L5, L5/L6, L6/L7, L7/S1 intervertebral disc space
  - Osteolytic mixed with osteoproliferative changes at L7 caudal end-plate
  - Marked osteolytic changes at S1 cranial end-plate

- Diagnosis or differential diagnosis
  - L7/S1 discospondylitits: main diagnosis
  - Spondylosis deformans at L3/L4, L4/L5, L5/L6, L6/L7, L7/S1: likely to be incidental findings; its clinical significance is uncertain


Questions?